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A PROJECT REPORT ON

"Wi-Fi Traffic Anomaly Detection Using Isolation Forest Algorithm"

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In

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Submitted by

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CERTIFICATE

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DECLARATION

We hereby declare that the project work entitled **"Wi-Fi Traffic Anomaly Detection Using Isolation Forest Algorithm"** submitted to the **Visvesvaraya Technological University, Belagavi** during the academic year **2026-2027**, is record of an original work done by us under the guidance of **Mrs. Deepti N.N ,** Assistant Professor, Department of Computer Science and Engineering Rajiv Gandhi Institute of Technology , Bengaluru and this project work is submitted in the partial fulfillment of requirements for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering.** The results embodied in this project have not been submitted to any other University or Institute for award of any degree or diploma.

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ABSTRACT

With the rapid expansion of wireless communication technologies, Wi-Fi networks have become an essential part of modern digital infrastructure. However, the increasing usage of these networks has also led to a rise in security threats such as unauthorized access, abnormal traffic behaviour, and cyber-attacks. Traditional intrusion detection systems primarily rely on signature-based or rule-based techniques, which are ineffective in identifying unknown or zero-day attacks. This creates a need for intelligent and adaptive systems capable of detecting anomalies in real time.

This project proposes a Wi-Fi traffic anomaly detection system using the Isolation Forest Algorithm, which is an efficient unsupervised machine learning technique. The algorithm works by isolating data points through random partitioning, making it particularly suitable for detecting anomalies that significantly differ from normal traffic patterns. Unlike supervised learning models, it does not require labelled datasets, making it more practical for real-world applications where labeled data is limited or unavailable. The system collects real-time Wi-Fi traffic parameters such as packet size and transmission duration, pre-processes the data using normalization techniques, and feeds it into the trained model for prediction. A user-friendly interface is developed using Streamlit to visualize the results. The dashboard displays live traffic logs, highlights anomalies in red, and presents graphical representations to help users easily identify unusual patterns in network activity.

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INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 Overview

Wi-Fi networks have become an essential part of modern communication systems, providing wireless connectivity for computers, mobile devices, applications, and other network-enabled systems. The continuous exchange of packets over a Wi-Fi network generates a large amount of network traffic that can vary depending on the type of communication and network activity. Monitoring this traffic is important for understanding network behaviour and identifying activities that may indicate abnormal or potentially harmful communication.

The proposed system, Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest**,** is designed to monitor live network traffic and identify anomalous traffic patterns using machine learning. The system captures network packets from the active Wi-Fi interface and processes the captured information to obtain the characteristics required for classification. These characteristics are represented using 68 network traffic features**,** which are processed using the proposed system scaler before being supplied to the trained proposed system Random Forest model.

The Random Forest model analyses the processed traffic features and classifies the observed network activity as normal or anomalous. The classification results are integrated with a real-time monitoring dashboard, enabling the user to observe live network traffic statistics, traffic sources, and detected anomalies. This provides a convenient way to monitor network activity without requiring continuous manual inspection of individual packets. The system combines packet capture, traffic feature processing, machine learning based classification, and real-time visualization into a single monitoring framework.

1.1.1 Outdoor cases

Wi-Fi networks are increasingly used in outdoor environments such as public areas, educational campuses, transportation facilities, commercial spaces, and other locations where wireless connectivity

is required over a wider area. In such environments, network traffic can vary significantly because multiple users and devices may connect to the same wireless infrastructure at different times. This variation makes continuous monitoring important for identifying unusual traffic behaviour. Outdoor Wi-Fi environments may contain a large number of connected devices communicating through shared wireless access points. Changes in user activity, device connectivity, network load, and communication patterns can result in significant variations in network traffic. Monitoring such traffic manually can be difficult, particularly when the network is active for long periods. The proposed system can be applied as a monitoring mechanism for such outdoor Wi-Fi environments by capturing live network traffic through the available Wi-Fi interface and processing the captured traffic using the proposed system Random Forest model. The system extracts the required 68 traffic features and uses the V6 scaler before performing classification. The resulting classification is displayed through the real-time dashboard, allowing abnormal traffic patterns to be observed during network operation. The use of machine learning based anomaly detection in outdoor Wi-Fi environments can support continuous observation of network behaviour and assist in identifying traffic that differs from expected patterns. The dashboard provides a centralized view of live traffic statistics, traffic sources, and detected anomalies, making the monitoring process more accessible to the user.

Thus, outdoor Wi-Fi environments represent a practical application area for real-time network traffic monitoring and anomaly detection, where automated analysis can help in observing changing traffic conditions and identifying potentially anomalous network activity.

1.1.2 Indoor cases

Indoor Wi-Fi networks are commonly used in environments such as homes, offices, educational institutions, laboratories, hospitals, and commercial buildings. These environments may contain multiple wireless devices that continuously exchange data through one or more access points. The resulting network traffic changes according to the number of connected users, applications being accessed, and communication activities taking place within the network.

In an indoor environment, network traffic may include normal activities such as web browsing, file transfers, application communication, and other routine data exchanges. Continuous monitoring of this traffic is useful for identifying patterns that differ from normal network.

1.2 Motivation

The increasing dependence on Wi-Fi networks for communication and data exchange has created a need for effective monitoring of wireless network traffic. As the number of connected devices and network-based applications continues to increase, Wi-Fi networks generate large and continuously changing volumes of traffic. Identifying abnormal traffic patterns from such continuous network activity can be difficult through manual observation alone.

Traditional network monitoring approaches generally require continuous observation of traffic information and may depend on predefined rules or manual analysis. Such approaches can become difficult to maintain when network traffic changes frequently. This creates a need for an automated mechanism that can analyse network traffic and identify patterns that differ from normal behaviour.The motivation for the proposed system is to apply machine learning to real-time Wi-Fi network traffic monitoring. The proposed system Random Forest model provides a classification-based approach in which captured traffic is processed into 68 features and passed through the proposed system scaler before classification. This allows the system to analyse incoming traffic and identify whether the observed traffic is normal or anomalous.

1.3 Problem Identification

Wi-Fi networks continuously generate network traffic from multiple connected devices and applications. As the volume and variety of traffic increase, identifying abnormal network behaviour from live traffic becomes difficult through manual monitoring. Individual packets and traffic records contain a large amount of information, making continuous inspection time-consuming and difficult to perform consistently. A major problem is the difficulty of distinguishing normal network activity from anomalous traffic patterns in real time. Conventional monitoring approaches may require predefined rules or manual analysis and may not provide an effective mechanism for automatically classifying changing traffic behaviour. In addition, continuously generated network traffic requires a monitoring mechanism that can process traffic information and provide detection results without requiring manual examination of every traffic record.

1.4 Scope of the project

The scope of the project is to develop a real-time system for monitoring Wi-Fi network traffic and detecting anomalous traffic behaviour using a machine learning based classification approach. The system focuses on capturing live network traffic from the active Wi-Fi interface, processing the captured information, and providing classification results through a real-time monitoring dashboard. The proposed system uses the proposed system Random Forest model for traffic classification. The captured network traffic is processed into 68 features, which are passed through the proposed system scaler before being provided to the trained Random Forest model. Based on the learned traffic patterns, the model classifies the observed traffic as normal or anomalous.

The scope of the project includes the following activities:

- Live Wi-Fi Traffic Capture: Capturing network packets from the active Wi-Fi interface using the available packet capture mechanism.
- Traffic Feature Processing: Extracting and processing the required 68 network traffic features from the captured traffic.
- Feature Scaling: Applying the proposed system scaler to transform the extracted features into the representation expected by the trained model.
- Random Forest Classification: Using the trained proposed system Random Forest model to classify network traffic and identify anomalous traffic patterns.
- Real-Time Monitoring: Continuously processing incoming traffic and updating the monitoring information during network operation.
- Dashboard Visualization: Displaying live traffic statistics, traffic sources, live network traffic, and detected anomalies through the proposed system dashboard.
- Demonstration Attack Mode: Providing a controlled demonstration mechanism for observing the behaviour of the anomaly detection system using prepared traffic samples.

1.5 Objective and Methodology

The primary objective of the project is to develop a real-time Wi-Fi network traffic anomaly detection system using the proposed system Random Forest model. The system is designed to monitor live network traffic, process the captured traffic information, classify the observed traffic, and present the detection results through a real-time dashboard.

The specific objectives of the project are:

- To capture live network traffic from the active Wi-Fi interface.
- To process the captured traffic and obtain the required 68 network traffic features.
- To apply the proposed system scaler to the extracted features before model prediction.
- To use the trained proposed system Random Forest model for classifying network traffic.
- To identify traffic patterns classified as anomalous by the trained model.
- To provide real-time visualization of traffic statistics, traffic sources, and detected anomalies.
- To provide a controlled demonstration mode for observing the anomaly detection behaviour.
- To provide an automated and centralized mechanism for monitoring Wi-Fi network traffic.

Methodology:

The methodology of the proposed system consists of a sequence of stages that connect live packet capture with machine learning based anomaly classification. The major stages are as follows:

1. Wi-Fi Interface Selection:
   The system operates on the active Wi-Fi network interface through which live network traffic is available.
2. Live Packet Capture:

Network packets are captured continuously from the Wi-Fi interface using the packet capture mechanism available in the system.

3. Traffic Processing:
   The captured packets are processed to obtain the information required for network traffic analysis.

4. Feature Extraction:
   The relevant traffic characteristics are represented using 68 features required by the proposed system Random Forest model.

5. Feature Scaling:
   The extracted feature values are processed using the proposed system scaler so that the input is transformed into the expected format for model prediction.

1.6 Existing System

Existing Wi-Fi network monitoring systems generally rely on conventional network monitoring tools, rule-based detection mechanisms, or manual inspection of captured network traffic. These systems are mainly used to observe network activity, identify connected devices, monitor bandwidth usage, and analyse packet-level information. Traditional approaches often depend on predefined rules or signatures to identify known forms of abnormal network behaviour. While such approaches can be effective for previously identified patterns, they may have limitations when network traffic changes dynamically or when an unusual pattern does not match an existing rule. Manual inspection of network packets can also become difficult because Wi-Fi networks continuously generate a large volume of traffic.

Some existing systems provide packet capture and analysis capabilities, but the user may need to inspect the captured information separately using network analysis tools. This can make continuous real-time anomaly identification and interpretation more difficult. In addition, conventional monitoring tools may focus primarily on displaying network information rather than combining live traffic processing with machine-learning-based classification.

1.7 Proposed System

The proposed system is a Real-Time Wi-Fi Network Traffic Anomaly Detection System using the V6 Random Forest model. The system is designed to monitor Wi-Fi network traffic continuously and identify abnormal traffic patterns in real time. Unlike conventional monitoring approaches that mainly depend on predefined rules or manual analysis, the proposed system uses a machine-learning-based approach to classify network traffic automatically. The system captures live network traffic through the available Wi-Fi interface and processes the captured traffic to obtain 68 network traffic features. These features are prepared in the same format used during the development of the proposed system model. Before classification, the extracted features are processed using the proposed system scaler to maintain consistency with the trained model.

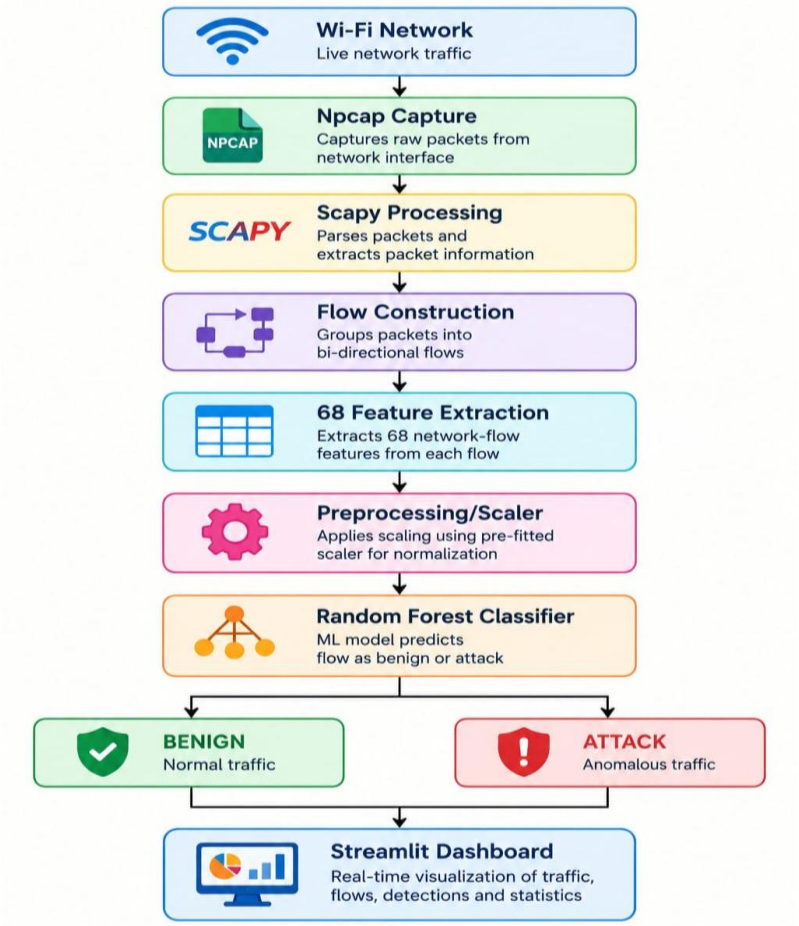


Figure 1.1 System Architecture

Block diagram:

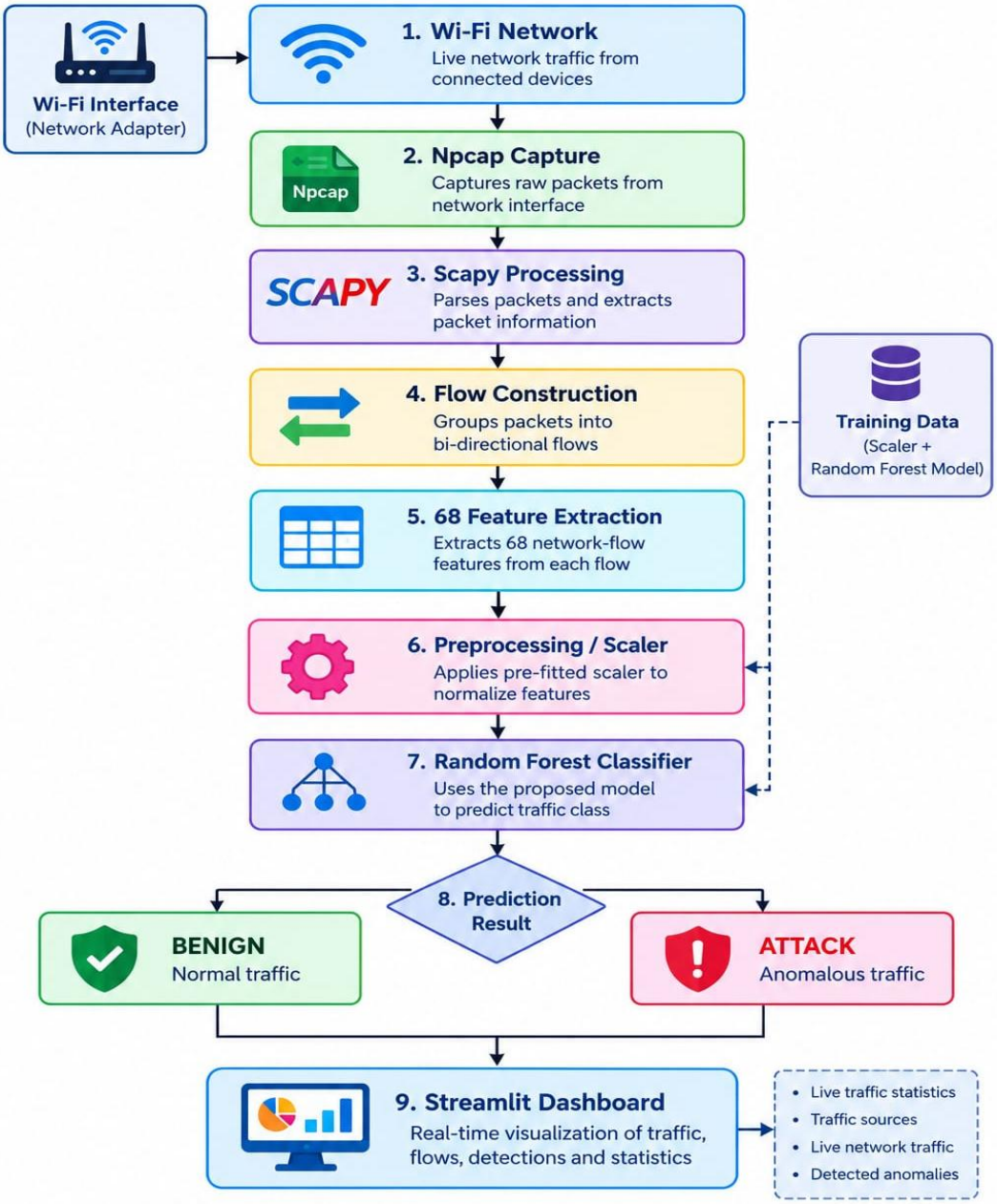


Figure 1.2 Workflow Diagram

1.8 Outcome of the project

The outcome of the project is the development of a real-time Wi-Fi network traffic anomaly detection system using the proposed system Random Forest model. The system successfully combines live network traffic monitoring with machine-learning-based anomaly classification to provide an automated method for identifying unusual network behavior. The developed system captures live Wi-Fi traffic and processes the collected information into 68 network traffic features. These features are passed through the proposed system scaler and then analyzed by the proposed system Random Forest classifier**.** Based on the learned traffic patterns, the system identifies whether the observed traffic is normal or anomalous.

1.9 Report Organization

Chapter 1:

Provides an overview of Wi-Fi traffic anomaly detection. It discusses the motivation, problem identification, scope, objectives, methodology, existing system, proposed system, expected outcomes, and summary of the project.

Chapter 2:

Presents the study of existing research and related works on Wi-Fi traffic monitoring, network anomaly detection, machine learning techniques, and the Isolation Forest algorithm.

Chapter 3:

This chapter discusses the analysis of the proposed system, including feasibility studies such as technical, economic, operational, schedule, and social feasibility.

Chapter 4:

Chapter describes the functional and non-functional requirements of the system. It also presents the required software, hardware, and other system requirements.

Chapter 5:

Explains the overall architecture and design of the proposed Wi-Fi anomaly detection system. It includes the system workflow, data flow, system diagrams, and design characteristics.

Chapter 6:

This chapter describes the implementation of the proposed system using the Isolation Forest algorithm. It explains the different modules, data processing, feature extraction, anomaly detection, and dashboard implementation.

Chapter 7:

Presents the testing and validation of the system. Different test cases are used to verify the functionality and accuracy of the anomaly detection system.

Chapter 8:

This chapter presents the screenshots and sample outputs of the implemented system, including Wi-Fi traffic information, normal traffic, and detected anomalies.

1.10 Introduction Summary

Wi-Fi networks have become an essential part of modern communication and provide connectivity across indoor and outdoor environments. The continuous increase in connected devices generates a large amount of network traffic, making manual monitoring and identification of abnormal activities difficult. Therefore, an automated approach is required to monitor network traffic and identify unusual patterns effectively. The proposed project, Real-Time Wi-Fi Network Traffic Anomaly Detection Using proposed system Random Forest**,** addresses this requirement by combining live Wi-Fi traffic monitoring with machine-learning-based anomaly detection.

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

2.1 Related Work

Network security has become an important research area due to the continuous growth of computer networks, wireless communication, and Internet-based applications. The increasing volume of network traffic makes manual inspection difficult and creates a requirement for automated techniques capable of identifying abnormal or potentially malicious traffic. Intrusion Detection Systems (IDS) are commonly used to monitor network activity and identify suspicious behaviour. Existing approaches can broadly be categorized into signature-based and anomaly-based detection. Signature-based systems identify previously known attack patterns using predefined rules, whereas anomaly-based systems attempt to identify traffic that differs from learned or expected behaviour. Machine learning has become an important approach for anomaly-based network security because it can learn relationships between traffic characteristics and classification outcomes.

Several studies have investigated machine learning algorithms including Decision Trees, Support Vector Machines, Naïve Bayes, Random Forest, neural networks, and deep learning models for network intrusion detection. Research surveys indicate that the choice of algorithm depends on factors such as dataset characteristics, detection requirements, training complexity, prediction time, and the intended deployment environment.

The proposed project builds on this research direction by applying a trained Random Forest model to live Wi-Fi network traffic. Instead of analyzing only stored datasets, the V6 implementation connects traffic capture, feature processing, machine learning prediction, and real-time dashboard visualization.

Paper Title	Authors	Year	Methodologies	Algorithm	limitations
Deep Reinforcement Learning-Driven Dynamic Spectrum Access in Dense Wi-Fi Environments	A . Alauthman & T.Shra'a	2025	DRL-based dynamic channel selection	DQN, PPO	High computational Cost
Real-Time IoT Based Wi-Fi Anomaly Detection	Patel et al.	2023	Real-time traffic monitoring	Isolation Forest	Limited scalability
Hybrid ML Model for Wi-Fi Security	Gupta & Rao	2021	Hybrid supervised + unsupervised learning	Isolation Forest +SVM	Complex parameter tuning
Deep Learning Based Wireless Anomaly Detection	Ahmed et al	2020	Neural - network reconstruction	Autoencoder	High computational complexity
Wi-Fi Intrusion Detection Based Random Forest	Kumar & Singh	2019	Ensemble learning with feature ranking	Random Forest	Require extensive features

Network Intrusion Detection Using Isolation Forest	Li & Wang	2018	Unsupervised anomaly detection	Isolation Forest	High false positives
Anomaly Detection in Wireless Networks Using ML	Sharma et al.	2016	Supervised classification	SVM	Poor zero-day detection

Table 2.1 LITERATURE SURVEY

2.2 Machine Learning Based Network Anomaly Detection

Machine learning based network anomaly detection is an important area of research in network security and intrusion detection. The primary purpose of anomaly detection is to identify network behaviour that differs significantly from expected or normal communication patterns. Traditional network monitoring systems generally depend on predefined rules, thresholds, or signatures. Although such approaches can be effective for detecting known conditions, they may have difficulty identifying previously unseen or changing network behaviours. Machine learning provides an alternative approach by learning relationships and patterns from network traffic data.

Network traffic contains a large amount of information that can be used for analysis. A network flow or traffic record may include information such as source and destination addresses, source and destination ports, communication protocols, packet counts, byte counts, packet lengths, connection duration, packet arrival patterns, and statistical properties of the communication.

Instead of examining these parameters individually, machine learning algorithms can analyse combinations of multiple characteristics and determine whether a particular traffic instance resembles normal or anomalous behaviour.

Machine learning approaches for network anomaly detection can generally be divided into supervised, unsupervised, and hybrid approaches. Supervised learning methods are trained using labelled network traffic in which traffic records are associated with predefined classes, such as normal or attack traffic. Instead, the algorithm attempts to identify patterns, clusters, outliers, or deviations within the available traffic.

2.3 Random Forest for Network Traffic Classification

Random Forest is a supervised ensemble machine learning algorithm based on multiple decision trees. The algorithm was introduced by Leo Breiman and combines several decision-tree predictors rather than relying on a single tree. During model construction, randomized feature selection and training procedures are used to produce a collection of different trees. The final prediction is obtained by combining the decisions generated by the individual trees. This ensemble approach can improve robustness compared with relying on one decision tree.

A decision tree makes predictions by applying a sequence of feature-based decisions. In network traffic classification, a tree may examine characteristics such as packet counts, durations, byte values, or other extracted traffic features. However, a single tree may become highly dependent on particular training examples. Random Forest addresses this limitation by constructing multiple trees and combining their predictions. Network traffic classification is particularly suitable for an ensemble approach because traffic behaviour is usually influenced by multiple characteristics simultaneously. The relationship between network features may not always be represented by a single simple decision rule. Random Forest can consider different combinations of features through its collection of decision trees and use their collective prediction to determine the traffic class.

The V6 architecture can therefore be represented as:

Live Wi-Fi Traffic → Feature Extraction → Preprocessing → Random Forest Model → Classification Result

This makes Random Forest the primary analytical component of the proposed real-time Wi-Fi anomaly detection system.

2.4 Wi-Fi and Wireless Network Traffic Monitoring

Wi-Fi networks differ from traditional wired networks because communication takes place through a shared wireless medium. Multiple devices may simultaneously use the same wireless network for different purposes, including web browsing, file transfers, streaming, application communication, software updates, and background services. Consequently, the characteristics of Wi-Fi traffic can change significantly depending on the connected devices and the activities being performed.

Wireless traffic is therefore dynamic rather than static. The amount of traffic, packet frequency, communication destinations, protocols, and connection behaviour may change over time. A monitoring system must be able to observe these changes while distinguishing legitimate variations from potentially anomalous behaviour.

The proposed V6 system addresses this requirement by operating on the active Wi-Fi interface and processing live network traffic. The captured traffic is transformed into the feature representation expected by the trained V6 Random Forest model. The resulting classification information is then supplied to the monitoring dashboard.

The wireless monitoring component therefore acts as the input layer of the complete system:

Wi-Fi Interface → Live Traffic Capture → Traffic Processing → Feature Extraction → Machine Learning Detection

2.5 Network Traffic Feature Extraction

Feature extraction is one of the most important stages in a machine learning based network anomaly detection system. Raw network packets contain detailed information about individual communications, but machine learning models generally require structured numerical input. Therefore, relevant characteristics must be extracted from network traffic and organised into a consistent feature vector.

The purpose of feature extraction is to transform complex network activity into measurable characteristics that can be processed by the machine learning model. Depending on the detection system, these characteristics may include packet-related statistics, byte counts, communication duration, protocol information, packet lengths, traffic rates, and other statistical properties.

The feature extraction process can be represented as:

Raw Network Traffic → Traffic Measurements → Feature Calculation → 68-Feature Vector

The quality of this stage directly influences the subsequent preprocessing and classification stages. Incorrect, missing, inconsistent, or incorrectly ordered features can result in unreliable predictions even when the trained Random Forest model itself is functioning correctly.

2.6 Feature Scaling in Machine Learning Based Detection

Feature scaling is an important preprocessing operation in many machine learning systems. Network traffic features can have significantly different numerical ranges. For example, one feature may contain relatively small numerical values while another may represent packet counts, byte counts, or duration values with much larger magnitudes.The preprocessing method used during model development must be reproduced during inference. If the model was trained using transformed or scaled values, applying raw values directly during live detection can create a mismatch between the training and deployment environments.

Consistent preprocessing is therefore important for maintaining model compatibility. The same transformation mechanism that was used when preparing the training data should be applied to newly captured traffic before classification.

In the proposed V6 system, the preprocessing scaler is stored separately as:

scaler_v6.pkl

The live detection pipeline extracts the required 68 features and applies the V6 scaler before passing the processed feature vector to the Random Forest classifier. This separation allows the preprocessing configuration to remain consistent with the trained model.

The processing sequence can be represented as:

Live Traffic → Feature Extraction → 68 Features → V6 Scaling → Random Forest Prediction

Separating the scaler from the model also provides a clearer system architecture. The feature extraction stage is responsible for generating the correct traffic representation, the scaler is responsible for applying the required preprocessing transformation, and the Random Forest model performs the final classification. Using the same preprocessing configuration during both training and live inference reduces the possibility of inconsistencies between the model development and deployment stages.

2.7 Machine Learning Based Intrusion and Anomaly Detection

Intrusion detection systems have evolved from traditional rule-based and signature-based approaches toward statistical and machine-learning-based methods. Signature-based detection generally searches for known patterns associated with previously identified attacks. This approach can be effective when attack signatures are known, but it may have limitations when previously unseen or modified behaviour occur. Machine learning based intrusion detection research has evaluated a broad range of algorithms. These include Decision Trees, Random Forest, Support Vector Machines, Naïve Bayes, artificial neural networks, deep learning techniques, and other classification approaches. Research commonly evaluates such algorithms using datasets such as KDD, NSL-KDD, CICIDS2017, and UNSW-NB15. However, algorithm selection for a real-time system should not be based solely on classification performance.

Random Forest was selected for the proposed V6 implementation because it provides an ensemble classification mechanism that is suitable for structured traffic features. This distinction is important because continuous model training would introduce additional computational requirements and could make the real-time detection process more complex. Instead, the V6 system separates the model development stage from the live detection stage.

The overall approach can therefore be expressed as:

Training Stage → Trained V6 Model

followed by:

Live Traffic → Feature Processing → Trained V6 Model → Detection Result

The project consequently focuses on applying an established machine learning classification method to live network traffic rather than proposing a new machine learning algorithm.

2.8 Real-Time Network Anomaly Detection Systems

Many intrusion detection experiments are performed using previously collected network datasets. Offline datasets are valuable because they allow researchers to train, test, compare, and evaluate different machine learning algorithms under controlled conditions. However, a real-world network monitoring system must operate on traffic as it occurs.

The major stages can be represented as:

Traffic Capture → Traffic Processing → Feature Generation → Preprocessing → Prediction → Result Recording → Visualization

Real-time operation introduces additional technical requirements. Network traffic can arrive continuously and may vary considerably in volume. If processing cannot keep pace with incoming traffic, the monitoring system may experience delays or lose useful information. Therefore, processing efficiency and prediction speed are important considerations in addition to classification performance. The proposed V6 system follows this real-time processing model. The live detector captures traffic from the Wi-Fi interface and maintains the information necessary for traffic analysis. The real-time approach also makes the system more suitable for demonstration and monitoring because the user can observe traffic activity and corresponding detection results while the network is operating.

2.9 Network Monitoring Dashboards and Visualization

The output of an anomaly detection system should be presented in a form that allows users to understand the current state of network activity. Raw classification records alone may not provide an efficient way to monitor continuously changing traffic. A dashboard can provide a centralized

representation of network activity by displaying traffic statistics, traffic sources, live traffic information, and detected anomalies. Visualization also makes it easier to observe changes in network behaviour over time.

The V6 system provides a dedicated real-time monitoring interface containing sections for:

- Current Wi-Fi Network
- Live Traffic Statistics
- Traffic Sources
- Live Network Traffic
- Detected Anomalies
- Demo Attack Mode

2.10 Comparison of Existing Approaches

Existing network anomaly detection approaches differ in terms of detection technique, adaptability, computational requirements, and deployment method.

Approach	Technique	Application	Advantage	Limitation
Rule-based detection	Predefined rules	Network monitoring	Simple to interpret	Limited adaptability
Statistical detection	Statistical analysis	Traffic monitoring	Detects deviations	Sensitive to changing traffic
Supervised ML	Classification models	Intrusion detection	Learns traffic patterns	Requires labelled training data
Random Forest	Ensemble classification	Traffic anomaly detection	Robust classification	Model size can be relatively large
Real-time ML monitoring	Packet/flow analysis + ML	Live network security	Continuous detection	Requires continuous processing

Table 2.2 Comparison of Existing Approaches

SYSTEM ANALYSIS

CHAPTER 3

SYSTEM ANALYSIS

3.1 Introduction to System Analysis

System analysis is the process of examining the requirements, functions, components, and operational workflow of a proposed system before its implementation. It helps in understanding the problem being addressed, identifying the limitations of existing approaches, and defining the requirements necessary to develop an effective solution. For the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system, the analysis focuses on the continuous monitoring of Wi-Fi traffic and the automated identification of abnormal network behaviour. The system is designed to capture live network traffic, extract 68 network traffic features, apply the proposed system scaler, and classify the processed traffic using the proposed system Random Forest model.

Thus, system analysis provides the foundation for defining the functional requirements, system workflow, hardware and software requirements, and design considerations needed to implement the proposed Wi-Fi traffic anomaly detection system effectively.

3.2 Feasibility Study

A feasibility study is conducted to determine whether the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system can be developed and operated effectively within the available resources and requirements. The study evaluates the technical, operational, and economic feasibility of the proposed system.

3.2.1. Technical Feasibility

Technical feasibility evaluates whether the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system can be successfully developed using the available hardware, software, and technical resources.

The proposed system is technically feasible because it uses a standard computing environment with a suitable Wi-Fi interface for capturing network traffic. The captured traffic is processed to obtain 68 network traffic features, which are passed through the proposed system scaler to maintain the required feature representation before classification.

The processed features are then supplied to the proposed system Random Forest model for identifying normal and anomalous traffic. Since the trained model and supporting feature and scaler files are already available, real-time detection does not require model retraining during operation. The system also provides **a** real-time dashboard for displaying network traffic information and detected anomalies. A controlled/demo attack mode is available for demonstrating and testing the anomaly detection workflow using prepared traffic samples.

Component	**Requirement / Purpose**
Operating System	Windows-based environment suitable for the implemented system
Programming Language	Python
Network Interface	Active Wi-Fi interface
Packet Capture	Mechanism for capturing live network traffic
Feature Processing	Processing of 68 traffic features
Feature Scaling	V6 scaler
Classification Model	V6 Random Forest
Model File	random_forest_v6.pkl
Scaler File	scaler_v6.pkl
Feature Information	features_v6.pkl
Feature Importance	feature_importance_v6.csv
Result Storage	Live detection result file
User Interface	V6 real-time monitoring dashboard

Table 3.1 Technical Feasibility

3.2.2. Economic Feasibility

Economic feasibility determines whether the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system can be developed and operated without requiring excessive financial resources. The proposed system is economically feasible because it mainly depends on software-based components and commonly available computing resources. The trained proposed system Random Forest model**,** proposed system scaler, and feature configuration can be used for real-time prediction without requiring additional model-training infrastructure during operation.

Resource	**Requirement**	**Cost Consideration**
Computer/Laptop	Used for system development and execution	Standard existing hardware can be used
Wi-Fi Connectivity	Required for capturing live Wi-Fi network traffic	Existing network infrastructure can be utilized
Python Environment	Required for development and execution	Software-based and freely available
Machine Learning Libraries	Required for feature processing and Random Forest prediction	Open-source software libraries
Packet Capture Support	Required for live network traffic capture	Software/network component
V6 Random Forest Model	Used for traffic classification and anomaly detection	Existing trained model
V6 Scaler	Used for preprocessing the 68 features	Existing trained component
Real-Time Dashboard	Used for monitoring and displaying results	Developed as part of the project
Storage	Required for storing model and result files	Standard system storage is sufficient

Table 3.2 Economic Feasibility

3.2.3 Operational Feasibility

Operational feasibility determines whether the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system can be effectively used in a practical network monitoring environment. The proposed system is operationally feasible because it provides an automated workflow for monitoring Wi-Fi traffic and identifying abnormal network behaviour. Live traffic is captured and processed continuously, reducing the need for manual inspection of individual packets. The extracted 68 network traffic features are processed using the proposed system scaler and classified by the proposed system Random Forest model**.**

The system presents the classification results through a real-time monitoring dashboard**,** allowing users to observe traffic activity and detected anomalies in a centralized interface. This makes the monitoring process easier to understand and supports timely observation of unusual traffic behaviour. The controlled/demo attack mode also supports operational testing by allowing the system's anomaly detection process to be demonstrated using prepared traffic samples. Therefore, the proposed system can be practically operated for real-time Wi-Fi traffic monitoring and anomaly detection with minimal manual intervention.

Operation	**System Activity**
Network Monitoring	Monitors the active Wi-Fi network
Packet Capture	Captures available live network traffic
Traffic Processing	Processes the captured network traffic
Feature Extraction	Generates the required 68 traffic features
Preprocessing	Applies the V6 scaler to the extracted features
Classification	Uses the V6 Random Forest model for traffic classification
Anomaly Detection	Identifies traffic classified as anomalous
Result Recording	Records the detection results
Visualization	Displays monitoring information through the V6 dashboard

Table 3.3 Operational Feasibility

3.2.4 Schedule Feasibility

Schedule feasibility evaluates whether the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system can be completed within the planned project duration. The project is divided into a sequence of manageable activities so that each stage can be completed systematically without affecting the subsequent stages.

3.2.5 Social Feasibility

Social feasibility determines whether the proposed system provides practical benefits and can be used effectively by its intended users.Wi-Fi networks are widely used in educational institutions, offices, homes, commercial environments, and other locations. Continuous network activity can make manual traffic monitoring difficult.

Benefit	Description
Automated Monitoring	Reduces the need for continuous manual inspection of Wi-Fi network traffic.
Real-Time Observation	Allows users to observe current network traffic and monitoring information.
Anomaly Identification	Helps identify traffic classified as anomalous by the V6 Random Forest model.
Centralized Visualization	Provides network monitoring information through a single dashboard.
Easy Interpretation	Presents traffic statistics and detected anomalies in an organized format.
Demonstration Support	Demo Attack Mode provides a controlled way to demonstrate the anomaly detection capability.
Improved Monitoring Support	Assists users in observing network activity and identifying potentially unusual traffic.

Table 3.4 Social benefits of the proposed system

REQUIREMENT ANALYSIS

CHAPTER 4

REQUIREMENT ANALYSIS

4.1 Functional Requirements

Functional requirements define the essential operations that the proposed system must perform. The Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system is designed to capture live network traffic, process the captured packets, classify traffic using the V6 Random Forest model, and display the results through a real-time monitoring dashboard.

The main functional requirements of the system are described below:

1. Wi-Fi Network Monitoring

The system must monitor the active Wi-Fi network and observe available network traffic. It should provide information about the currently monitored network and support continuous traffic observation.

2. Live Packet Capture

The system must capture live network packets through the selected Wi-Fi network interface. Packet capture provides the input required for subsequent traffic processing and anomaly detection.

3. Traffic Processing and Feature Extraction

The captured network traffic must be processed to obtain the required features for classification. The V6 system uses a set of 68 features to represent the relevant characteristics of network traffic.

4. Feature Scaling

The extracted features must be processed using the V6 scaler before being supplied to the classification model. This ensures that the input data is transformed using the preprocessing component associated with the trained model

5. Random Forest Classification

The system must use the trained V6 Random Forest model to classify the processed network traffic. The model uses the extracted and scaled features to generate traffic classification results.

6. Anomaly Detection

The system must identify traffic classified as anomalous by the V6 Random Forest model. The detected results should be made available for monitoring and further inspection.

7. Real-Time Dashboard

The system must display monitoring information through the V6 dashboard. The interface includes the current Wi-Fi network, live traffic statistics, traffic sources, live network traffic, detected anomalies, and Demo Attack Mode.

8. Result Recording

The system should record live detection results so that the classification output can be reviewed after processing.

9. Demo Attack Mode

The system must provide a demonstration mode to illustrate the anomaly detection process using the available demonstration data and the V6 classification model.

4.2 Non-Functional Requirements

Non-functional requirements specify the quality characteristics and operational expectations of the system. These requirements focus on performance, reliability, usability, maintainability, and the system's ability to support continuous network monitoring.

1. Performance

The system should process captured network traffic and generate classification results with minimum unnecessary delay. The dashboard should present available traffic information and detected anomalies in a timely manner.

2. Reliability

The system should operate consistently during network monitoring. The packet capture, feature processing, scaling, and classification components should work together to provide the intended detection functionality.

3. Usability

The dashboard should be simple and understandable. The user should be able to observe network statistics, traffic sources, live traffic, and detected anomalies without needing to examine raw packet information manually.

4. Real-Time Operation

The system should support continuous observation of available Wi-Fi traffic and update the monitoring interface with processed information as it becomes available.

5. Maintainability

The implementation should allow the individual source files and supporting model components to be maintained when required. The V6 frontend, live detection components, scaler, and trained model should remain associated with the correct version of the system.

6. Compatibility

The system should operate in the configured Python environment and work with a supported Wi-Fi interface and packet capture setup.

7. Security

The system should handle captured traffic and detection results responsibly. Access to network traffic should be limited to authorized monitoring activities, and the system should be used in accordance with applicable network policies.

4.3 Software Requirements

The software requirements define the programming environment, libraries, model files, and supporting components needed to develop and execute the proposed system. Python is used for system implementation, while the packet capture and machine learning components support live traffic monitoring and classification.

Category	Software / Component	Purpose
Programming Language	Python	Development of traffic processing, machine learning, and monitoring components
Machine Learning	Scikit-learn	Loading and executing the Random Forest model
Data Processing	Pandas	Processing and handling traffic data and detection results
Numerical Processing	NumPy	Numerical operations during feature processing
Packet Capture	Scapy / Npcap	Capturing live network packets
Classification Model	V6 Random Forest	Classification of network traffic
Model File	random_forest_v6.pkl	Stores the trained V6 Random Forest model
Scaler File	scaler_v6.pkl	Performs feature scaling before prediction
Feature File	features_v6.pkl	Stores the feature information required by the V6 model

Category	Software / Component	Purpose
Feature Importance	feature_importance_v6.csv	Stores feature-importance information of the V6 model
Frontend	frontend_v6.py	Provides the V6 real-time monitoring dashboard
Live Detection	live_detection_v6.py	Supports live detection processing
Live Detector	live_detector_v6.py	Supports live network traffic detection

Table 4.1 Primary software requirements

4.3.1 Operating System

The proposed system is implemented for a Windows-based environment with a compatible Wi-Fi network interface and packet capture support. Npcap is required for the configured packet capture setup, while Python provides the environment for executing the detection and dashboard components. The computer must have a working Wi-Fi adapter and appropriate driver support to access the network interface. The required Python libraries and V6 model files must also be available in the configured environment. The exact operating system version and Python version should be documented according to the versions installed and used during project testing.

4.4 Hardware Requirements

The hardware requirements describe the computing resources and network components needed to operate the proposed system. A computer with adequate processing capability and memory is required to execute the Python application, process network traffic, and load the trained Random Forest model. The system also requires an active Wi-Fi interface that supports the configured packet capture mechanism. Standard input and display devices are sufficient for operating the application and viewing the monitoring dashboard.

Hardware Component	Minimum Requirement	Purpose
Processor	Intel Core i5 / AMD Ryzen 5 or equivalent	Running the application and ML processing
RAM	8 GB or above	Supporting live traffic processing and model execution
Storage	256 GB or above	Storing software, model files, and system data
Wi-Fi Adapter	Compatible Wi-Fi adapter	Capturing live Wi-Fi network traffic
Network Connection	Active Wi-Fi connection	Providing network traffic for monitoring
Display	Standard monitor/display	Viewing the real-time dashboard
Keyboard and Mouse	Standard	System operation and interaction

Table 4.2 Hardware Requirements

4.4.1 Requirement Analysis Summary

The requirement analysis identifies the functional, non-functional, software, and hardware requirements of the Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system. The functional requirements establish the main operations, including live packet capture, processing of 68 features, feature scaling, Random Forest classification, anomaly detection, and dashboard visualization. The non-functional requirements describe the expected characteristics of the system, including performance, reliability, usability, maintainability, compatibility, and responsible handling of captured traffic. The software requirements identify the Python environment, packet capture components, machine learning libraries, and V6 model files required for execution. The hardware requirements establish the need for a suitable computer and compatible Wi-Fi interface.

SYSTEM DESIGN

CHAPTER 5

SYSTEM DESIGN

The system design defines the overall structure, components, interactions, and processing flow of the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system. The purpose of the system design is to convert the requirements identified during the analysis phase into a practical software architecture that can capture live Wi-Fi traffic, process the captured traffic, extract meaningful network features, classify the traffic using the trained V6 Random Forest model, and present the results through a real-time monitoring interface.

5.1 System Design Overview

The proposed system is designed to operate as a real-time network monitoring and anomaly detection application. Unlike a system that analyses only previously collected network traffic, the proposed system is intended to continuously process traffic available through the Wi-Fi interface.

The system begins its operation when the monitoring process is started. The packet capture component obtains network packets and forwards them to the processing stage. The processing stage converts packet-level information into traffic information suitable for feature extraction. Feature extraction is an important stage because the machine learning model does not directly operate on raw packet information. Instead, the traffic is represented using a predefined set of 68 features. These features form the input vector expected by the V6 Random Forest model.

The complete system can therefore be divided into the following major stages:

Network Input → Packet Capture → Traffic Processing → Feature Extraction → Feature Scaling → Random Forest Classification → Anomaly Detection → Result Management → V6 Dashboard

This architecture provides a continuous pipeline from network traffic acquisition to anomaly visualization.

5.1.1 System Architecture Overview

The system architecture consists of several functional layers that work together to perform real-time anomaly detection.

Network Layer

The network layer represents the Wi-Fi environment from which the traffic is monitored. Network packets generated by communication between devices are available through the system's Wi-Fi interface. The network layer acts as the source of input data for the monitoring application. The system does not need to modify the network traffic; instead, it observes and processes the available traffic for analysis.

Packet Capture Layer

The packet capture layer is responsible for obtaining live packets from the Wi-Fi interface. The project uses a packet capture mechanism based on Scapy, with Npcap providing the required packet capture support in the Windows environment. The captured packets are passed to the next stage for processing. This layer provides the connection between the real network environment and the software-based detection pipeline.

Traffic Processing Layer

The traffic processing layer handles the captured packet information and prepares it for feature extraction. Raw packets contain information at the packet level, while machine learning classification requires structured traffic information. This stage therefore organizes and processes the available packet information so that meaningful traffic characteristics can be extracted.

Feature Extraction Layer

The feature extraction layer generates the features required by the trained machine learning model. The V6 system uses 68 traffic features as the model input. The extracted feature vector must maintain the required structure and ordering expected by the trained V6 model. This ensures that each feature is provided to the correct model input position.

Preprocessing Layer

The preprocessing layer contains the V6 scaler used to transform the extracted features before classification. The scaler is an important part of the trained model pipeline because the same preprocessing approach used during model development must also be applied to live traffic. The resulting transformed feature vector is then passed to the classifier.

Machine Learning Layer

The machine learning layer contains the trained V6 Random Forest model. The model receives the processed feature vector and generates the corresponding classification output. Random Forest is suitable for the proposed system because it can work with multiple traffic characteristics and combine the decisions of multiple decision trees to produce a classification result.

Anomaly Detection Layer

The anomaly detection layer interprets the output produced by the machine learning model. The classification result is converted into detection information that can be presented to the user. This layer acts as the logical connection between machine learning classification and real-time monitoring.

Result Management Layer

The result management layer handles the generated monitoring and detection information. Results can be maintained so that the frontend can access and display the latest information.

Presentation Layer

The presentation layer consists of the V6 frontend dashboard. It provides the user with a visual representation of the monitoring process.

The dashboard can present information such as:

- Current Wi-Fi network information
- Live traffic statistics

- Traffic sources
- Live network traffic
- Detected anomalies
- Detection-related information
- Demo Attack Mode

The layered structure allows changes in one component to be made with minimal impact on other components.

5.1.2 System Flow Design

The system flow describes the sequence in which data moves through the proposed anomaly detection system. When the application is started, the monitoring component initializes the network monitoring process. The system then begins obtaining available network packets through the Wi-Fi interface.

The process is repeated continuously while monitoring is active.

Detailed System Flow

Step 1 – Start Monitoring

The user starts the V6 monitoring application through the frontend.

Step 2 – Capture Network Traffic

The system accesses the available Wi-Fi interface and captures network packets.

Step 3 – Process Captured Packets

The captured packets are processed to obtain traffic-level information.

Step 4 – Extract Features

The system extracts the required 68 network traffic features.

Step 5 – Scale Features

The extracted feature vector is transformed using the V6 scaler.

Step 6 – Classify Traffic

The scaled feature vector is supplied to the trained V6 Random Forest model.

Step 7 – Detect Anomaly

The model prediction is interpreted by the anomaly detection component.

Step 8 – Record Result

The generated detection information is made available to the result management component.

Step 9 – Display Result

The V6 frontend displays the latest traffic and anomaly information.

Step 10 – Continue Monitoring

The process continues for subsequent network traffic while real-time monitoring remains active.

5.1.3 Design Characteristics

The proposed system has several design characteristics that support its intended purpose of real-time Wi-Fi traffic anomaly detection.

Modularity

The system is divided into separate logical modules. Packet capture, traffic processing, feature extraction, scaling, classification, anomaly detection, result management, and visualization have separate responsibilities. This modularity makes the system easier to understand and maintain.

Real-Time Operation

The system is designed to process network traffic continuously rather than relying only on offline analysis. New traffic can pass through the detection pipeline as monitoring continues.

Machine Learning Integration

The system integrates a trained V6 Random Forest model directly into the detection pipeline. This allows the trained model to be used for classification of processed live network traffic.

Consistent Feature Processing

The live detection pipeline uses the same feature structure and associated preprocessing components expected by the V6 model. This reduces the possibility of inconsistencies between training-time and live-time processing.

Maintainability

The separation of the model, scaler, feature information, live detection components, and frontend improves maintainability. Individual components can be modified or improved without redesigning the entire system.

Extensibility

The architecture provides a foundation for future improvements. Additional features, updated models, enhanced visualization, or additional detection functionality can be incorporated into the system.

User-Oriented Design

The system presents complex network monitoring and machine learning results through a graphical interface. This makes the output easier to interpret than raw packet information or raw model predictions.

5.2 Gantt Chart

5.2.1 Gantt Chart for Phase One



Figure 5.1 Gantt Chart for Phase One

5.2.2 Gantt Chart for Phase Two

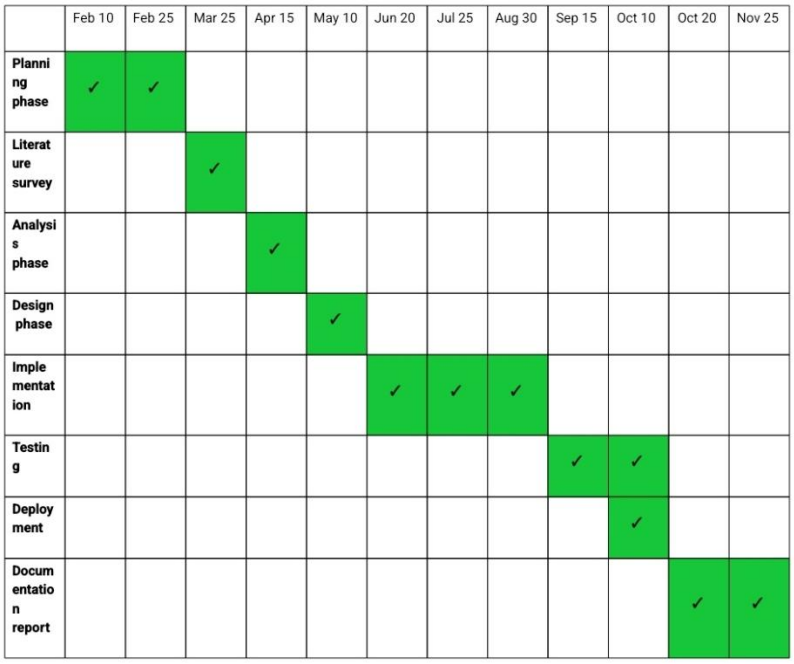


Figure 5.2 Gantt Chart for Phase Two

5.3 Life Cycle Model

The development of the Wi-Fi anomaly detection system follows an iterative life cycle model. An iterative model is appropriate for this project because the machine learning model, feature processing pipeline, live detection mechanism, and frontend require repeated testing and refinement.

The life cycle can be represented as:

Requirement Analysis → System Design → Development → Testing → Evaluation → Refinement → Integration → Final Testing

The iterative nature of the development process is particularly useful for a machine learning-based application because improvements can be introduced based on testing and evaluation results.

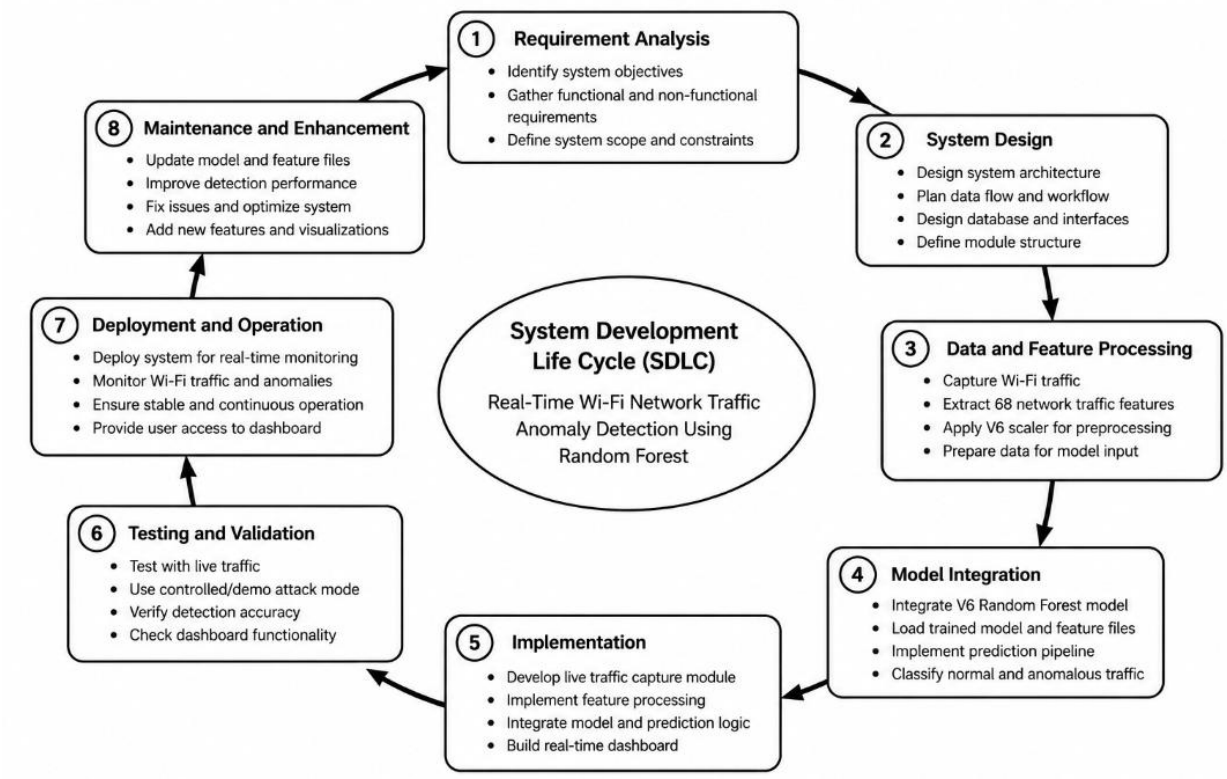


Figure 5.3 Life Cycle Mode

5.4 Data flow diagram

The below diagram represents the Data Flow Diagram (DFD) of the Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system. It is divided into Level 0, Level 1, and Level 2, where each level provides increasing detail about how network traffic moves through the system.

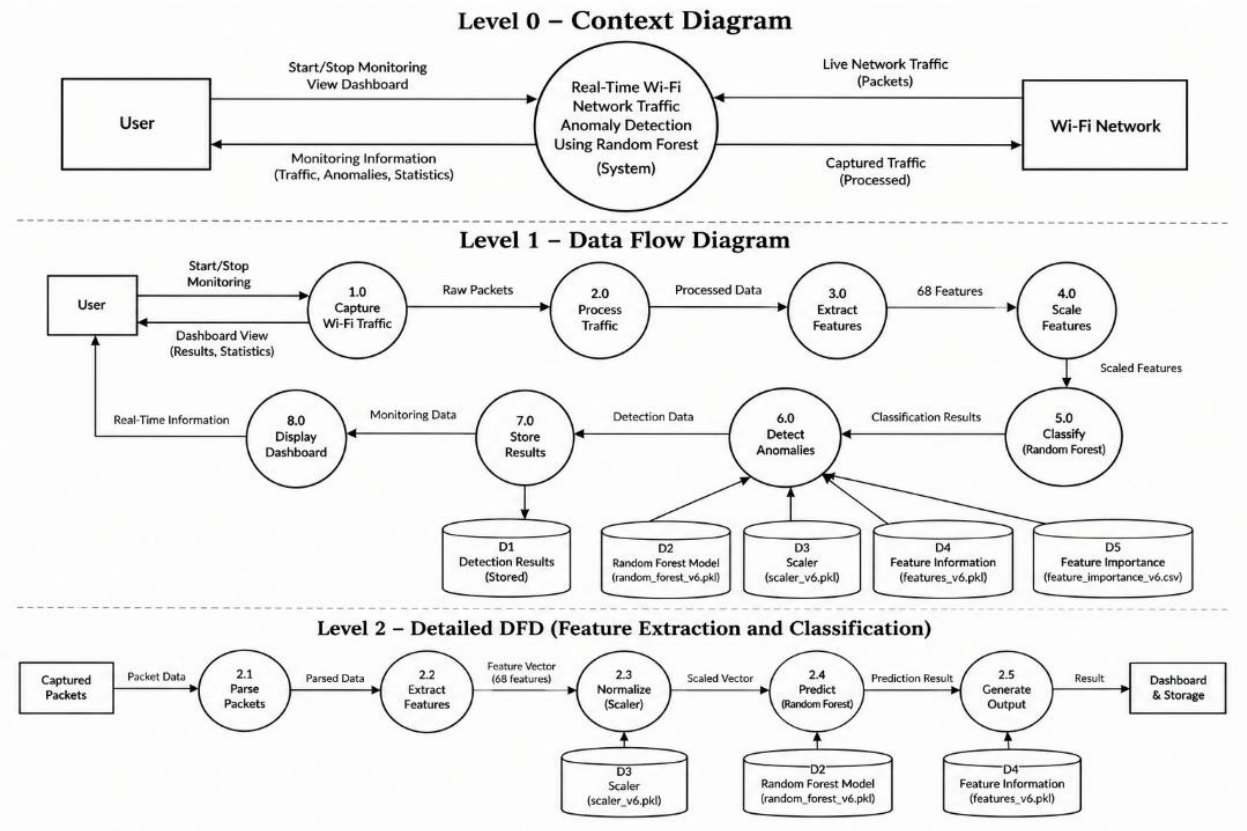


Figure 5.4 Data flow diagram

Process	Name	Explanation	Input	Output
1.0	Capture Wi-Fi Traffic	Captures live packets from the Wi-Fi network through the available network interface.	Live Wi-Fi traffic	Raw packets

Process	Name	Explanation	Input	Output
2.0	Process Traffic	Processes the captured raw packets and prepares them for feature extraction.	Raw packets	Processed data
3.0	Extract Features	Extracts the network traffic characteristics required by the machine learning model.	Processed data	68 features
4.0	Scale Features	Applies the trained scaler to transform the extracted features into the required format for the Random Forest model.	68 features	Scaled features
5.0	Classify (Random Forest)	Uses the trained V6 Random Forest model to classify the processed network traffic.	Scaled features	Classification results
6.0	Detect Anomalies	Interprets the classification result and identifies whether the observed traffic represents an anomaly.	Classification results	Detection data
7.0	Store Results	Stores the generated detection and monitoring information for further use by the system.	Detection data	Monitoring data
8.0	Display Dashboard	Presents the processed traffic, statistics, and anomaly information to the user through the V6 dashboard.	Monitoring data	Real-time information

Table 5.1 Data Flow

5.5 Use case diagram

It shows the interaction between the User, the System, and the major functions performed during real-time network monitoring and anomaly detection.

Element	Description
User	The user interacts with the system through the dashboard. The user can start or stop monitoring, view real-time information, trigger the demo attack, view detection results, and manage system settings.
System	Represents the Real-Time Wi-Fi Network Traffic Anomaly Detection System. It performs packet capture, feature extraction, scaling, Random Forest prediction, and anomaly detection.
System Boundary	The rectangular boundary represents all functions that belong to the proposed Wi-Fi anomaly detection system.

Table 5.2 Actors and System

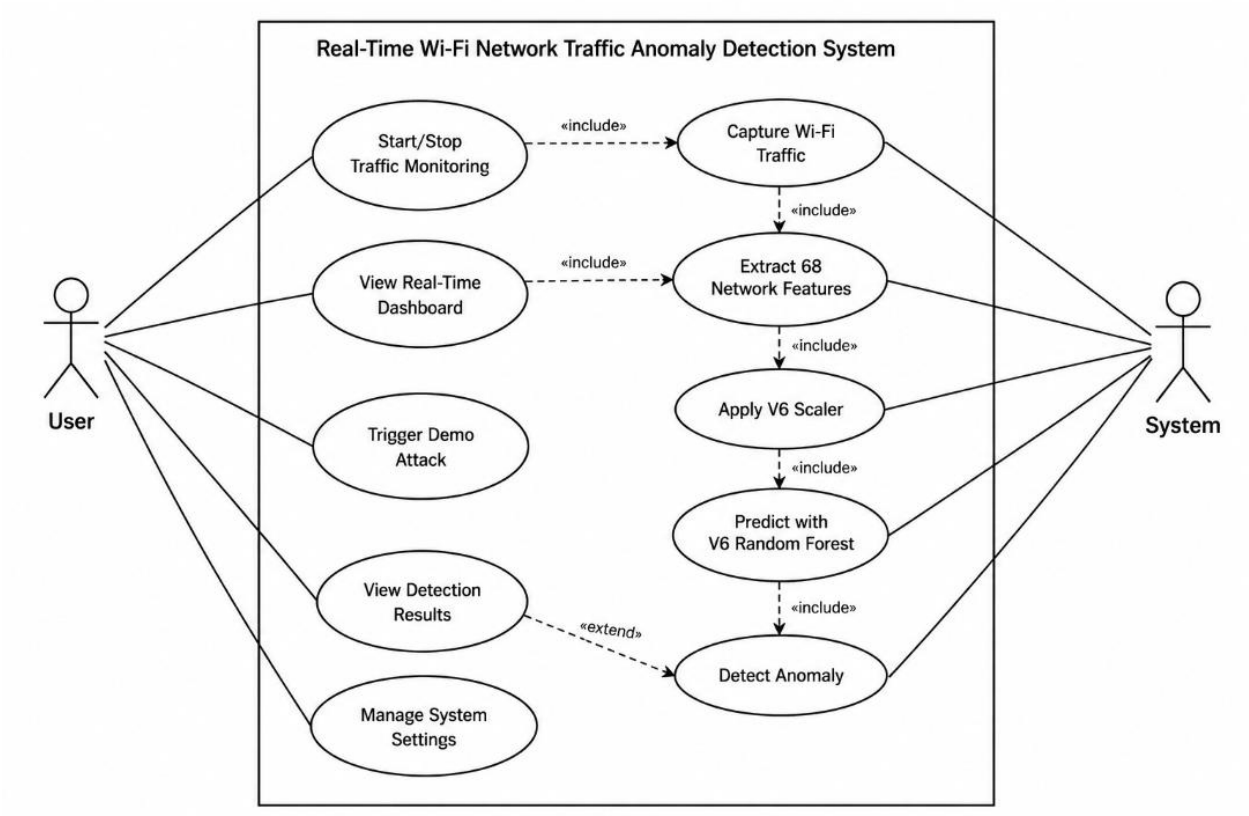


Figure 5.5 Use case Diagram

5.6 Sequence Diagram

The Sequence Diagram illustrates the sequence of interactions between the user and the different components of the Real-Time Wi-Fi Network Traffic Anomaly Detection System. It explains how a monitoring request is processed from the initial user action to the final display of the anomaly detection result.

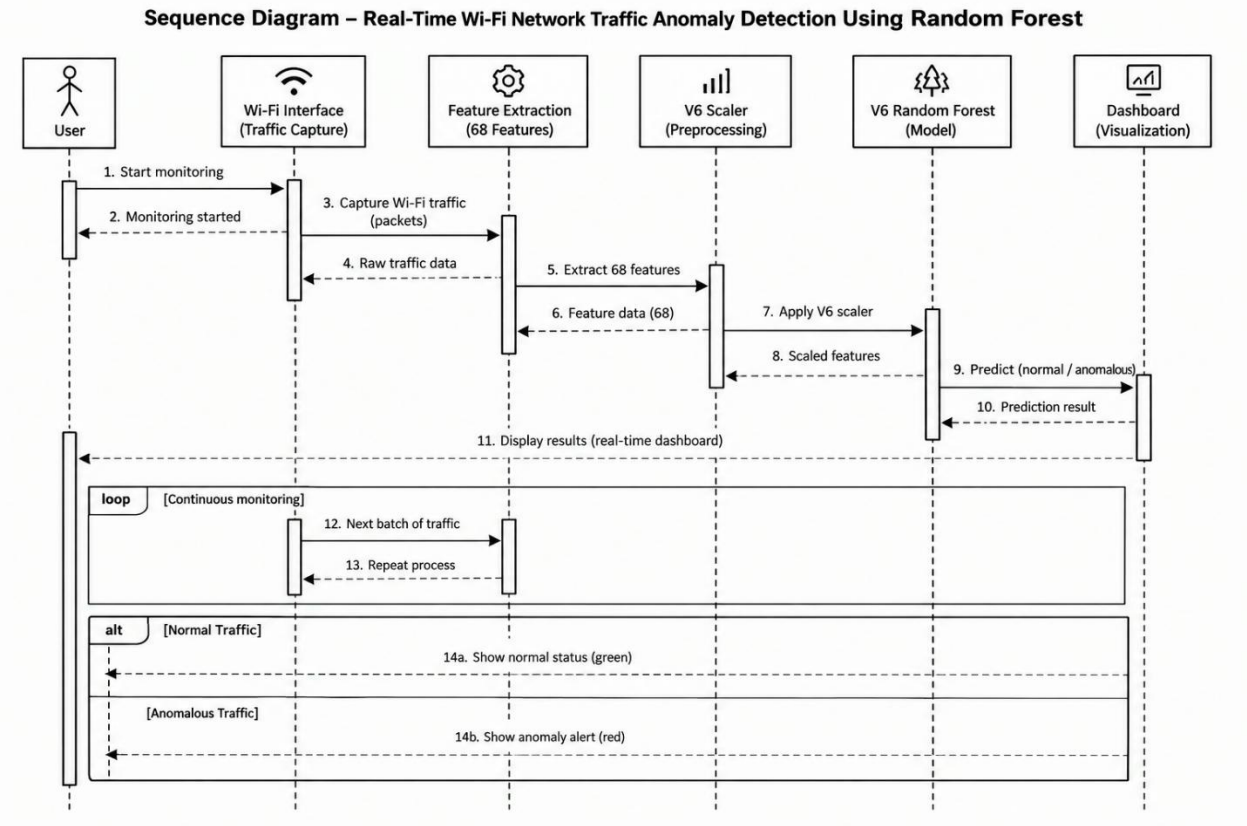


Figure 5.6 Sequence Diagram

Step	Participant	Interaction	Description
1	User → Wi-Fi Interface	Start monitoring	The user starts real-time traffic monitoring through the system.
2	Wi-Fi Interface → User	Monitoring started	The system confirms that monitoring has started.

Step	Participant	Interaction	Description
3	Wi-Fi Interface → Feature Extraction	Capture Wi-Fi traffic (packets)	Live Wi-Fi packets are captured and forwarded for analysis.
4	Feature Extraction → Wi-Fi Interface	Raw traffic data	The captured traffic information is made available for feature processing.
5	Feature Extraction → V6 Scaler	Extract 68 features	The system extracts the required 68 network traffic features from the captured traffic.
6	V6 Scaler → Feature Extraction	Feature data (68)	The extracted 68-feature vector is passed to the preprocessing stage.
7	V6 Scaler → V6 Random Forest	Apply V6 scaler	The V6 scaler transforms the extracted feature data into the required scaled format.
8	V6 Random Forest → V6 Scaler	Scaled features	The scaled feature vector is provided to the Random Forest model.
9	V6 Random Forest → Dashboard	Predict (normal/anomalous)	The V6 Random Forest model classifies the processed network traffic.
10	Dashboard → V6 Random Forest	Prediction result	The classification result is returned for presentation.
11	Dashboard → User	Display results	The dashboard displays the real-time traffic and detection results to the user.

Table 5.3 Sequence Diagram Table

5.7 Class Diagram

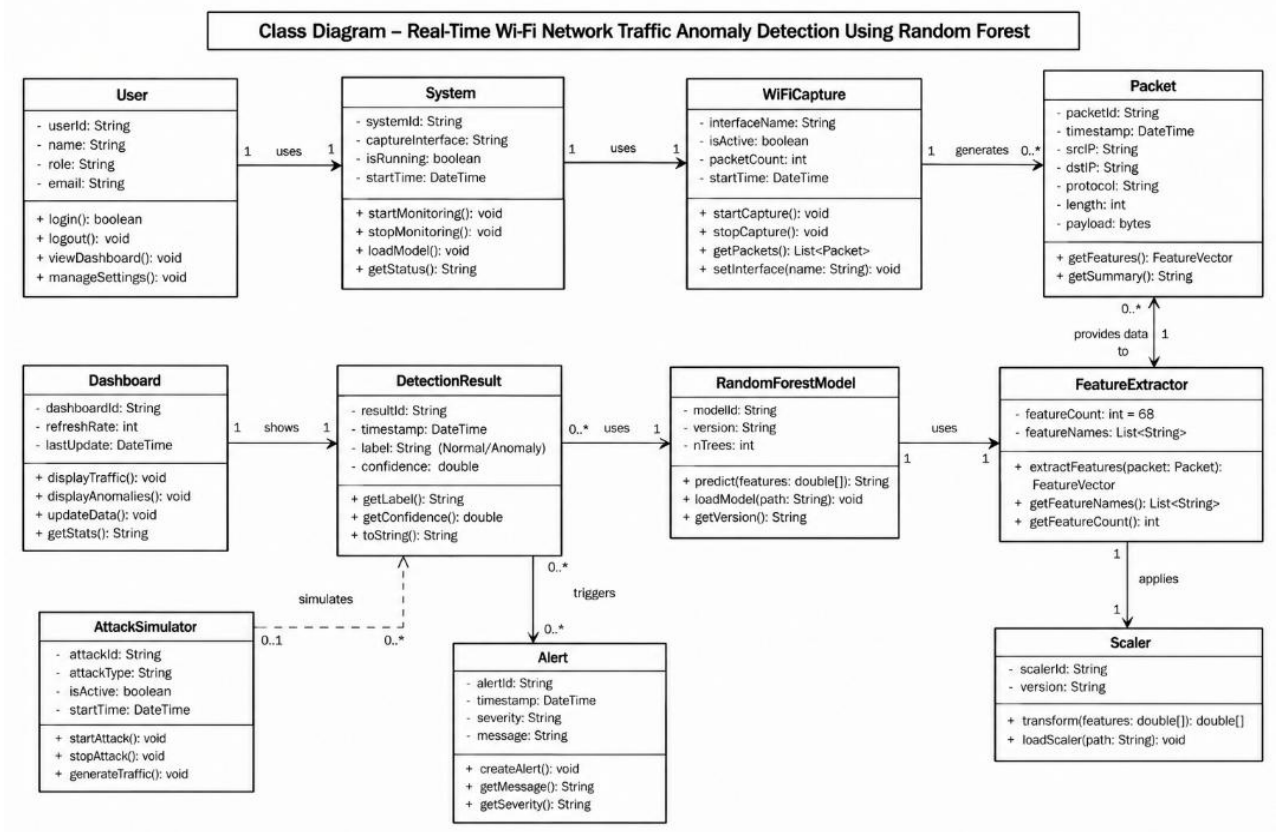


Figure 5.7 Class Diagram

The Class Diagram represents the logical structure of the Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system. It identifies the major classes or components of the system, their responsibilities, and the relationships between them. The diagram focuses on how the different modules work together to capture network traffic, process it, extract features, perform classification, detect anomalies, and display the results.

5.7.1 Relationships Between Classes

Source Class	Relationship	Target Class	Multiplicity	Explanation
User	Uses	System	1 : 1	A user interacts with the system to start monitoring, view the dashboard, and manage settings.
System	Uses	Wi-Fi Capture	1 : 1	The system uses the Wi-Fi capture component to obtain live network traffic.
Wi-Fi Capture	Generates	Packet	1 : 0..*	One Wi-Fi capture process can generate multiple network packets.
Packet	Provides data to	Feature Extractor	0..* : 1	Multiple captured packets can provide data to the feature extraction component.
Feature Extractor	Uses	Random Forest Model	1 : 1	The extracted feature information is ultimately used as input for model prediction.
Feature Extractor	Applies	Scaler	1 : 1	The extracted features are passed through the trained scaler before prediction.
Scaler	Provides processed data to	Random Forest Model	1 : 1	The scaled feature vector is provided to the Random Forest model.

Source Class	Relationship	Target Class	Multiplicity	Explanation
Random Forest Model	Produces	Detection Result	1 : 0..*	The model can produce multiple detection results during continuous monitoring.
Dashboard	Shows	Detection Result	1 : 1	The dashboard displays the generated detection results to the user.
Detection Result	Triggers	Alert	0..* : 0..*	Detection results can lead to the generation of alerts when anomalous traffic is identified.
Attack Simulator	Simulates	Detection Result	0..1 : 0..*	The attack simulator can generate traffic intended to produce detection results during demonstration or testing.

Table 5.4 Relationships Between Classes

SYSTEM IMPLEMENTATION

CHAPTER 6

SYSTEM IMPLEMENTATION

The system implementation phase converts the proposed design of the Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest into a functional software application. The implementation integrates live Wi-Fi packet capture, network-flow processing, feature extraction, feature preprocessing, Random Forest classification, result storage, and dashboard visualization.

The implemented system is divided into a backend live detection component and a frontend visualization component. The backend continuously captures network traffic from the selected Wi-Fi interface, organizes packets into flows, extracts the required 68 network traffic features, applies the preprocessing used by the V6 model, and performs classification using the trained V6 Random Forest model. The generated prediction, confidence, flow information, and timestamp are stored in a CSV result file.

The overall implementation can therefore be represented as:

Wi-Fi Interface → Packet Capture → Flow Formation → Feature Extraction → V6 Preprocessing → V6 Random Forest → Prediction → CSV Result Storage → Streamlit Dashboard

6.1 Modular Approach

The proposed system follows a modular approach in which different tasks are implemented as separate logical components. This approach makes the system easier to understand, test, maintain, and modify. Each module performs a specific operation and passes its output to the next stage of the detection pipeline.

6.1.1 Wi-Fi Packet Capture Module

The Wi-Fi packet capture module is responsible for obtaining live network packets from the selected Wi-Fi interface. The backend uses the Scapy packet-capture functionality and searches

the available network interfaces for the configured Wi-Fi interface. In the implementation, the interface is configured as "Wi-Fi" and the code also checks for the Intel Wi-Fi 6 AX101 adapter description.

A simplified portion of the implementation is:

```
INTERFACE_NAME = "Wi-Fi"


  for interface in conf.ifaces.values():
      name = getattr(interface, "name", "")
      description = getattr(interface, "description", "")


      if (
          name.lower() == INTERFACE_NAME.lower()
          or
          "Intel(R) Wi-Fi 6 AX101".lower()
          in description.lower()
      ):
          iface = interface
          break
```

The selected interface is subsequently supplied to the Scapy sniff() function for continuous packet capture.

6.1.2 Flow Management Module

Instead of immediately classifying every individual packet, the backend maintains network flows using a flow dictionary. The implementation maintains counters for captured packets, analyzed flows, predictions, attack results, and benign results.

The flow-management process identifies whether a packet belongs to an existing forward flow, an existing backward flow, or a new flow. The process_packet() function performs this operation.

The main processing logic is:

```
  if forward_key in flows:
      update_flow(
          flows[forward_key],
          packet,
          forward=True
      )
      return

  if backward_key in flows:
      update_flow(
          flows[backward_key],
          packet,
          forward=False
      )
      return
```

If the packet belongs to neither an existing forward nor backward flow, a new flow is created.

This flow-based approach allows the system to calculate statistical characteristics of network communication before performing classification.

6.1.3 Feature Extraction Module

The feature extraction module converts the captured flow information into the input representation required by the V6 Random Forest model.

The implementation contains a build_features() function that calculates traffic characteristics such as packet counts, packet lengths, inter-arrival times, packet rates, byte rates, header lengths, and other flow statistics.

The system specifically validates that the loaded V6 feature list contains **68 features**. If the feature

count is different, the implementation raises an error rather than continuing with an incompatible model input.

The generated feature dictionary includes network-flow characteristics such as:

- Destination port
- Flow duration
- Forward packet count
- Backward packet count
- Packet lengths
- Packet rates
- Inter-arrival times
- Header lengths
- Forward and backward statistics
- TCP flag information
- Flow-related statistical measurements

Thus, the feature extraction module forms the connection between raw network traffic and the machine learning model.

6.1.4 V6 Preprocessing Module

After the 68 features are generated, the backend converts them into a Pandas DataFrame using the exact feature order stored in the V6 feature file. Missing or invalid values are handled before the data is passed to the model.

The implementation first replaces infinite values with missing values and then fills missing values with zero. It subsequently applies the same log transformation used during V6 training.

The preprocessing sequence is:

68 Features → Invalid Value Handling → Missing Value Replacement → Log Transformation → V6 Scaling

A key implementation snippet is:

```
row = row.replace(
    [np.inf, -np.inf],
    np.nan
)
row = row.fillna(0)

row = (
    np.sign(row)
    *
    np.log1p(
        np.abs(row)
    )
)
```

The transformed data is then passed through the trained scaler:

```
X_scaled = scaler.transform(row)
```

This ensures that live network traffic is processed using the preprocessing configuration associated with the V6 model.

6.1.5 V6 Random Forest Classification Module

The Random Forest classification module contains the trained V6 machine learning model. The backend loads the model using joblib and verifies that the model file exists before starting live detection. The scaler and feature list are loaded in the same initialization stage.

The classification operation is performed after preprocessing:

```
prediction = model.predict(
    X_scaled
) [0]
```

```
probabilities = model.predict_proba(
    X_scaled
)[0]

confidence = float(
    np.max(probabilities)
)
```

The model output is converted into a human-readable label. A prediction value of 1 is represented as ATTACK, while the alternative result is represented as BENIGN. The corresponding confidence value is also calculated.

6.1.6 Result Storage Module

The result storage module records the output of the live detection process in a CSV file. The backend creates the required result directory if it does not already exist and initializes the result file with the required columns.

The stored information includes:

- Timestamp
- Source IP
- Source port
- Destination IP
- Destination port
- Protocol
- Flow duration
- Number of packets

6.2 Programming Code

The programming implementation consists mainly of the **V6 backend/live detection code** and the **V6 Streamlit frontend code**. The backend performs the actual network traffic analysis and machine learning prediction, while the frontend provides visualization and user interaction.

6.2.1 Backend Implementation

The backend imports the required libraries for file management, time processing, model loading, numerical computation, data processing, and network packet capture. Scapy is used for packet sniffing, while joblib is used to load the trained V6 model, scaler, and feature list.

Model and Scaler Loading

The V6 backend loads three important model-related files:

```
MODEL_PATH = r"..\model\v6\random_forest_v6.pkl"
SCALER_PATH = r"..\model\v6\scaler_v6.pkl"
FEATURE_PATH = r"..\model\v6\features_v6.pkl"
```

The model and preprocessing objects are loaded using joblib.load().

This separation ensures that the trained model, scaler, and feature configuration can be reused during real-time detection.

Feature Count Validation

The implementation verifies the number of features expected by the V6 model:

```
FEATURES = joblib.load(FEATURE_PATH)

if len(FEATURES) != 68:
    raise ValueError(
        f"Expected 68 features, found {len(FEATURES)}"
    )
```

This is an important validation step because the live feature vector must match the structure

expected by the trained Random Forest model.

6.2.2 Packet Capture Code

The backend uses Scapy's sniff() function to capture live traffic. The packet callback is connected to process_packet, while store=False prevents packets from being unnecessarily accumulated in Scapy's internal storage.

The implementation is:

```
sniff(
    iface=iface,
    prn=process_packet,
    store=False,
    timeout=1
)
```

The detector repeatedly executes this operation inside a loop and then checks for expired flows.

This forms the foundation of the real-time monitoring process.

6.2.3 Feature Extraction Code

The build_features() function generates the feature dictionary required by the model. It calculates flow duration, packet counts, packet rates, inter-arrival times, byte rates, packet-length statistics, and other traffic characteristics.

The generated feature dictionary is subsequently converted into a DataFrame using the V6 feature ordering.

```
row = pd.DataFrame(
    [features],
    columns=FEATURES
)
```

This ensures that the features are supplied to the model in the correct order.

6.2.4 Preprocessing and Prediction Code

The prediction function performs the complete machine-learning preprocessing and classification sequence.

The implementation:

```
row = row.replace(
    [np.inf, -np.inf],
    np.nan
)


row = row.fillna(0)
```

handles invalid and missing values. The same log transformation used during V6 training is then applied before scaling.

The scaler is applied using:

```
X_scaled = scaler.transform(row)
```

followed by Random Forest prediction:

```
prediction = model.predict(
    X_scaled
)[0]
```

The implementation also obtains class probabilities and calculates the maximum probability as the prediction confidence.

6.2.5 Attack and Benign Classification

The prediction is converted into the labels used by the application:

```
if prediction == 1:
    label = "ATTACK"
else:
    label = "BENIGN"
```

The system maintains separate counters for attack and benign classifications. This information is later used in the live detection summary and frontend dashboard.

6.2.6 Saving Detection Results

After prediction, the backend creates a Pandas DataFrame containing the flow information and classification result. The result is appended to the live detection CSV.

```
result.to_csv(
    RESULT_FILE,
    mode="a",
    header=False,
    index=False
)
```

This allows the frontend to continuously access newly generated detection results.

6.2.7 Frontend Implementation

The frontend is implemented using Streamlit and Pandas. It is responsible for presenting the backend results in a user-friendly dashboard.

6.2.8 Wi-Fi Status Code

The frontend obtains Wi-Fi information using the Windows networking command:

```
result = subprocess.run(
    ["netsh", "wlan", "show", "interfaces"],
    capture_output=True,
    text=True,
    encoding="utf-8",
    errors="ignore"
)
```

The resulting text is processed using regular expressions to extract SSID, signal, channel, band, receive rate, and transmit rate.

The extracted information is displayed using Streamlit metrics.

6.2.9 Live Result Loading

The frontend reads the backend-generated CSV using Pandas:

```
df = pd.read_csv(
    RESULT_FILE,
    names=columns,
    header=None
)
```

It then converts timestamps, confidence values, and packet counts to appropriate data types.

The dashboard limits live traffic to the most recent five minutes:

```
latest_time = df["Time"].max()
live_cutoff = latest_time - pd.Timedelta(minutes=5)
df = df[df["Time"] >= live_cutoff].copy()
```

This keeps the dashboard focused on current live traffic.

6.2.10 Live Traffic Statistics

The frontend calculates the number of total flows, benign flows, attack flows, and the anomaly rate.

```
total = len(df)
benign = int(
    (df["Prediction"] == "BENIGN").sum()
)
attack = int(
    (df["Prediction"] == "ATTACK").sum()
```

)

The anomaly rate is calculated from the number of detected attack flows relative to the total number of flows.

These values are displayed using Streamlit metric components.

6.2.11 Traffic Source Analysis

The frontend groups traffic based on source IP, destination IP, destination port, and protocol. It calculates the number of flows, attacks, and normal flows associated with each traffic source.

A status field is then generated:

" 🚨 ANOMALY"

if row["Attacks"] > 0

else "✅ NORMAL"

This allows the dashboard to highlight traffic sources associated with detected anomalies.

6.2.12 Anomaly Display

The dashboard filters the detection results to identify flows classified as ATTACK.

```
attacks = df[
    df["Prediction"] == "ATTACK"
].copy()
```

If no attack flows are found, the dashboard displays a message indicating that no anomalies have been detected. Otherwise, the detected anomalous flows are displayed in the anomaly section.

This provides a direct visual indication of the detection result to the user.

6.2.13 Demo Attack Mode

The frontend contains a controlled Demo Attack Mode implemented through a sidebar checkbox. When enabled, the dashboard loads previously processed attack flows from the V6 demonstration CSV file.

6.3 System Implementation Summary

The implementation of the Real-Time Wi-Fi Network Traffic Anomaly Detection System integrates network monitoring and machine learning classification into a complete application. The backend uses Scapy to capture live packets from the selected Wi-Fi interface and organizes the captured packets into network flows.

Final implementation flow

Wi-Fi Interface
↓
Scapy Packet Capture
↓
Flow Creation and Management
↓
68-Feature Extraction
↓
Data Cleaning + V6 Log Transformation
↓
V6 Scaler
↓
V6 Random Forest
↓
ATTACK / BENIGN + Confidence
↓
CSV Result Storage
↓
Streamlit V6 Dashboard
↓
Traffic Statistics + Anomaly Display

TESTING

CHAPTER 7

TESTING

Testing is an important phase in the development of the Real-Time Wi-Fi Network Traffic Anomaly Detection Using Random Forest system. The purpose of testing is to verify that the implemented system performs according to the requirements defined during the analysis and design phases. Testing also ensures that the different modules work correctly when integrated into a single real-time application.

The testing process for the proposed system covers the Wi-Fi packet capture module, traffic-flow processing, 68-feature extraction, V6 preprocessing, Random Forest classification, anomaly detection, result storage, and V6 dashboard. Both individual software components and the complete integrated system are considered during testing.

7.1 Validation and System Testing

Validation and system testing are performed to determine whether the implemented system satisfies its intended objectives. The testing process verifies both the correctness of individual software components and the behaviour of the complete real-time anomaly detection system.

The primary testing flow is:

Wi-Fi Traffic → Packet Capture → Flow Processing → 68 Features → V6 Preprocessing → Random Forest Prediction → Result Storage → Dashboard

Each stage is checked to ensure that the output generated by one component is suitable for the next component.

The major areas considered during validation and system testing include

- Wi-Fi interface detection
- Live packet capture
- Network-flow creation

- Feature extraction
- 68-feature validation
- Data preprocessing
- V6 scaler operation
- Random Forest prediction
- Attack/benign classification
- Confidence calculation
- CSV result generation

7.1.1 Software Testing

Software testing is performed to identify errors and verify that each software component operates according to its intended functionality. The testing is performed at different levels, including module-level testing, integration testing, and system-level testing.

A. Module Testing

Module testing verifies individual components separately before considering the complete application.

1. Wi-Fi Interface Testing

The Wi-Fi interface module is tested to verify that the application can identify the configured wireless network interface.

Test objective:
To verify that the system can locate and use the required Wi-Fi interface.

Expected result:
The application identifies the available Wi-Fi interface and starts packet capture successfully.

Result:
The module operates correctly when a valid Wi-Fi interface is available.

2. Packet Capture Testing

The packet capture module is tested using live network traffic.

Test objective:

To verify that network packets are successfully captured from the selected Wi-Fi interface.

Expected result:

Packets are continuously received by the backend while monitoring is active.

Result:

The packet capture module successfully receives live traffic and forwards packets for further processing.

3. Flow Processing Testing

The flow processing component is tested by generating multiple packets belonging to different network communications.

Test objective:

To verify that packets are correctly associated with forward and backward flows.

Expected result:

Packets belonging to the same communication are grouped into the appropriate flow.

Result:

The system successfully maintains active flows and processes completed flows after the configured timeout.

4. Feature Extraction Testing

The feature extraction module is tested using captured network-flow information.

Test objective:

To verify that the system generates the required network traffic features.

Expected result:

The system generates the required 68 features in the expected order.

Result:

The feature extraction module produces a feature vector compatible with the V6 preprocessing and Random Forest model.

The implementation also verifies the number of loaded features before performing prediction. If the feature count is not 68, the system raises a validation error rather than continuing with incompatible input.

5. Preprocessing and Scaler Testing

The preprocessing stage is tested using extracted feature values containing normal numerical values and potential invalid values.

Test objective:

To verify that the feature data is correctly cleaned and transformed before classification.

6. Random Forest Model Testing

The V6 Random Forest model is tested using valid scaled feature vectors.

Test objective:

To verify that the trained model accepts the processed input and produces a valid classification.

Expected result:

The model produces a valid prediction and corresponding confidence value.

Result:

The V6 Random Forest model successfully generates classification results for processed network flows.

7. Result Storage Testing

The result-storage module is tested after classification.

Test objective:

To verify that the prediction results are correctly written to the live detection CSV file.

B. Integration Testing

Integration	Testing Objective	Expected Result
Wi-Fi Interface → Packet Capture	Verify live packets are received	Packets are captured successfully
Packet Capture → Flow Processing	Verify captured packets are grouped into flows	Valid flows are generated
Flow Processing → Feature Extraction	Verify completed flows are converted into features	68 features are generated
Feature Extraction →	Verify feature vector is correctly	Valid processed feature

Integration	Testing Objective	Expected Result
Preprocessing	prepared	vector is generated
Preprocessing → Random Forest	Verify model receives valid input	Prediction is generated
Random Forest → Result Storage	Verify prediction is recorded	Detection result is saved
Result Storage → Dashboard	Verify frontend receives backend output	Results appear on dashboard
Dashboard → User	Verify information is clearly presented	User can view traffic and anomalies

Table 7.1 Integration Testing

7.1.2 Validation

Validation determines whether the developed system satisfies the intended requirements and objectives. While testing checks whether individual functions operate correctly, validation focuses on whether the overall system performs the intended task of real-time Wi-Fi network traffic anomaly detection.

The proposed system is validated at multiple levels.

Functional Validation

Functional validation verifies that the system provides the functions defined during the requirements analysis.

The system is validated for:

- Real-time Wi-Fi traffic capture.
- Network-flow processing.
- 68-feature extraction.
- V6 feature preprocessing.
- Random Forest classification.

- Attack and benign identification.
- Confidence calculation.
- Detection result storage.
- Real-time dashboard visualization.
- Anomaly display.
- Demo Attack Mode.

The implemented functions correspond to the intended system requirements.

Model Input Validation

The V6 model input is validated to ensure that the live feature vector matches the model configuration.

The system specifically checks that 68 features are available before classification. This prevents an incorrectly structured feature vector from being passed to the Random Forest model.

Output Validation

The output of the Random Forest model is validated before it is written to the result file.

The system generates a classification label and confidence value. The classification result is then stored together with the corresponding flow information.

7.2 Testing Summary

The testing phase verifies the functionality, integration, and overall behaviour of the proposed Real-Time Wi-Fi Network Traffic Anomaly Detection system. Individual modules such as Wi-Fi interface detection, packet capture, flow processing, feature extraction, preprocessing, Random Forest prediction, result storage, and dashboard visualization are tested independently.

Integration testing confirms that the output of one module is correctly passed to the next module. In particular, the connection between live packet capture, 68-feature extraction, V6 preprocessing, Random Forest classification, CSV result generation, and dashboard visualization is verified.

System testing then evaluates the complete application as a single integrated system.

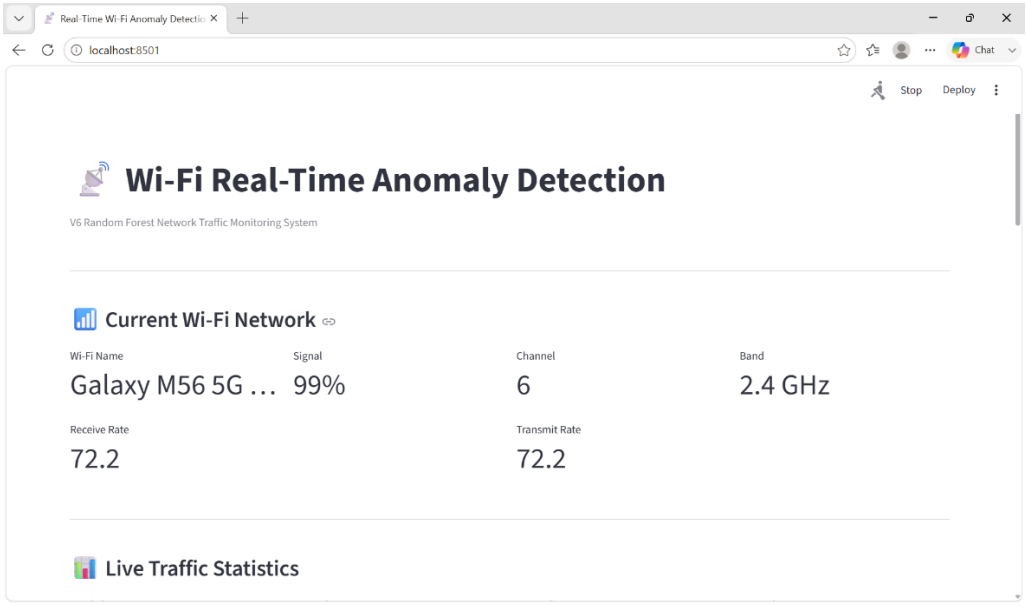
SAMPLE OUTPUT

CHAPTER 8

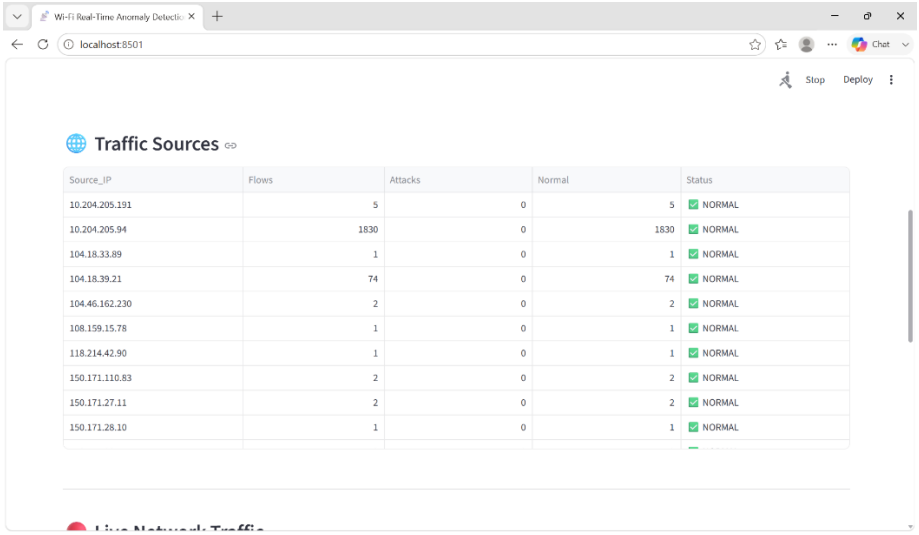
SAMPLE OUTPUT

8.1 Snapshots

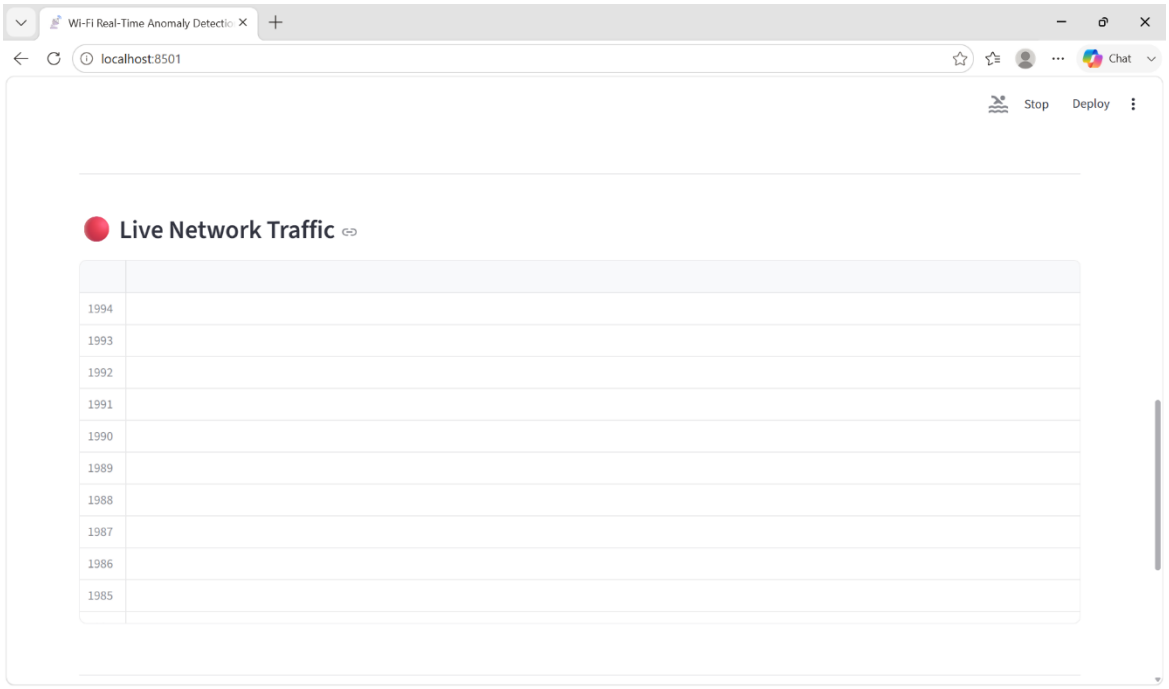
8.1.1 Current Wi-Fi Network Interface



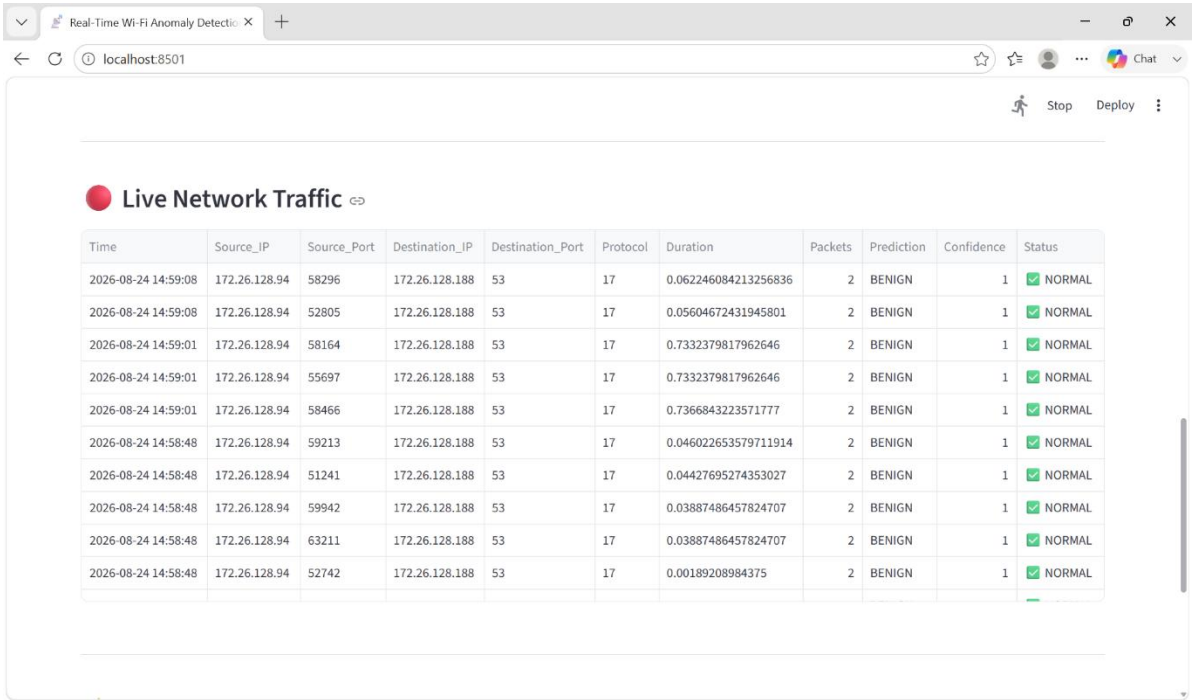
8.1.2 Traffic Sources



8.1.3 Live Network Traffic without data source

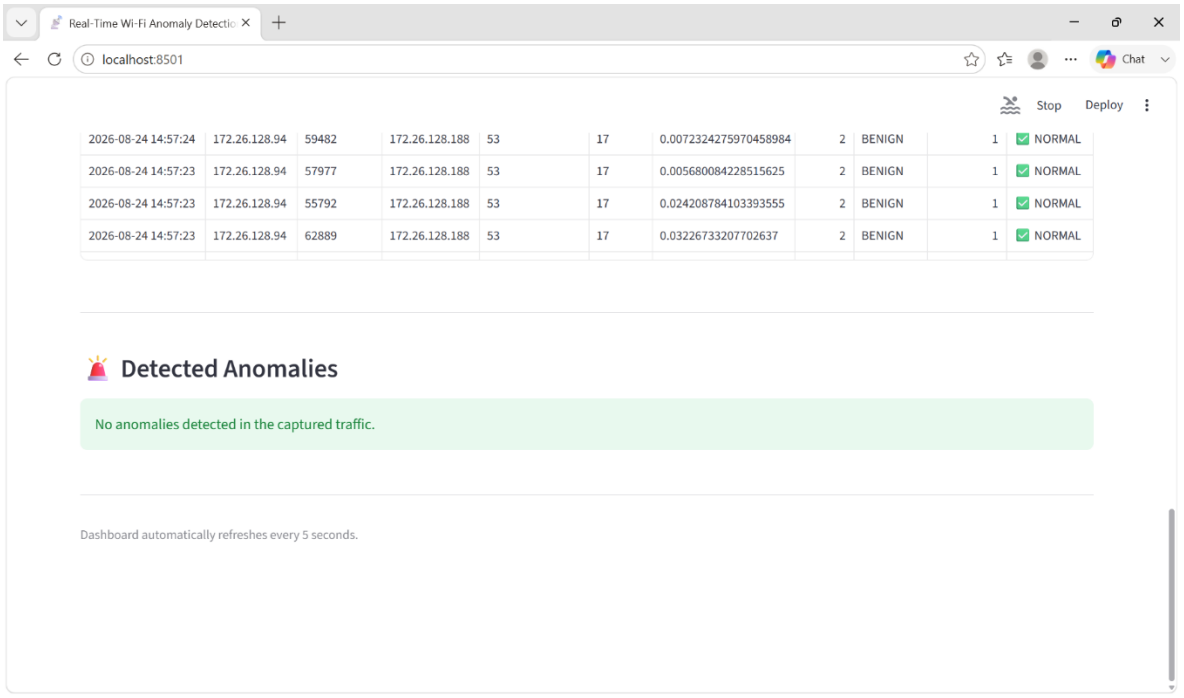


8.1.4 Live Network Traffic with data source

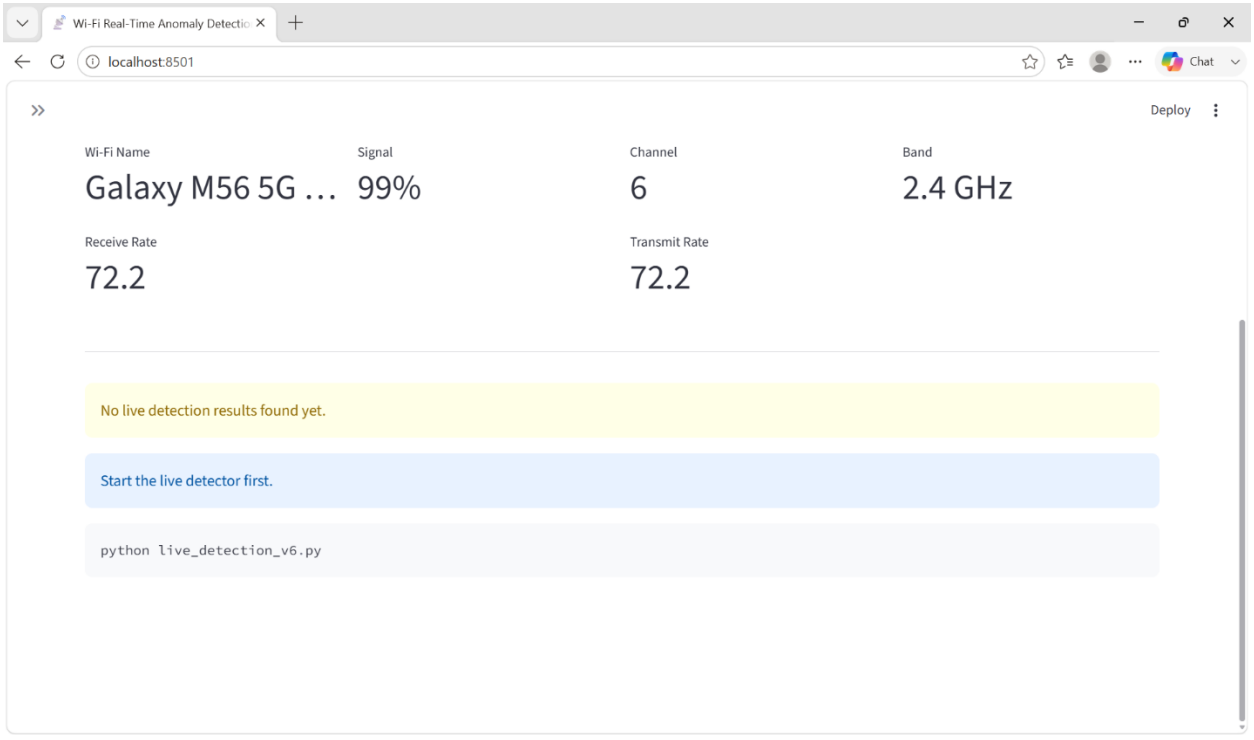


Time	Source_IP	Source_Port	Destination_IP	Destination_Port	Protocol	Duration	Packets	Prediction	Confidence	Status
2026-08-24 14:59:08	172.26.128.94	58296	172.26.128.188	53	17	0.062246084213256836	2	BENIGN	1	NORMAL
2026-08-24 14:59:08	172.26.128.94	52805	172.26.128.188	53	17	0.05604672431945801	2	BENIGN	1	NORMAL
2026-08-24 14:59:01	172.26.128.94	58164	172.26.128.188	53	17	0.7332379817962646	2	BENIGN	1	NORMAL
2026-08-24 14:59:01	172.26.128.94	55697	172.26.128.188	53	17	0.7332379817962646	2	BENIGN	1	NORMAL
2026-08-24 14:59:01	172.26.128.94	58466	172.26.128.188	53	17	0.7366843223571777	2	BENIGN	1	NORMAL
2026-08-24 14:58:48	172.26.128.94	59213	172.26.128.188	53	17	0.0460226535797119 14	2	BENIGN	1	NORMAL
2026-08-24 14:58:48	172.26.128.94	51241	172.26.128.188	53	17	0.04427695274353027	2	BENIGN	1	NORMAL
2026-08-24 14:58:48	172.26.128.94	59942	172.26.128.188	53	17	0.03887486457824707	2	BENIGN	1	NORMAL
2026-08-24 14:58:48	172.26.128.94	63211	172.26.128.188	53	17	0.03887486457824707	2	BENIGN	1	NORMAL
2026-08-24 14:58:48	172.26.128.94	52742	172.26.128.188	53	17	0.00189208984375	2	BENIGN	1	NORMAL

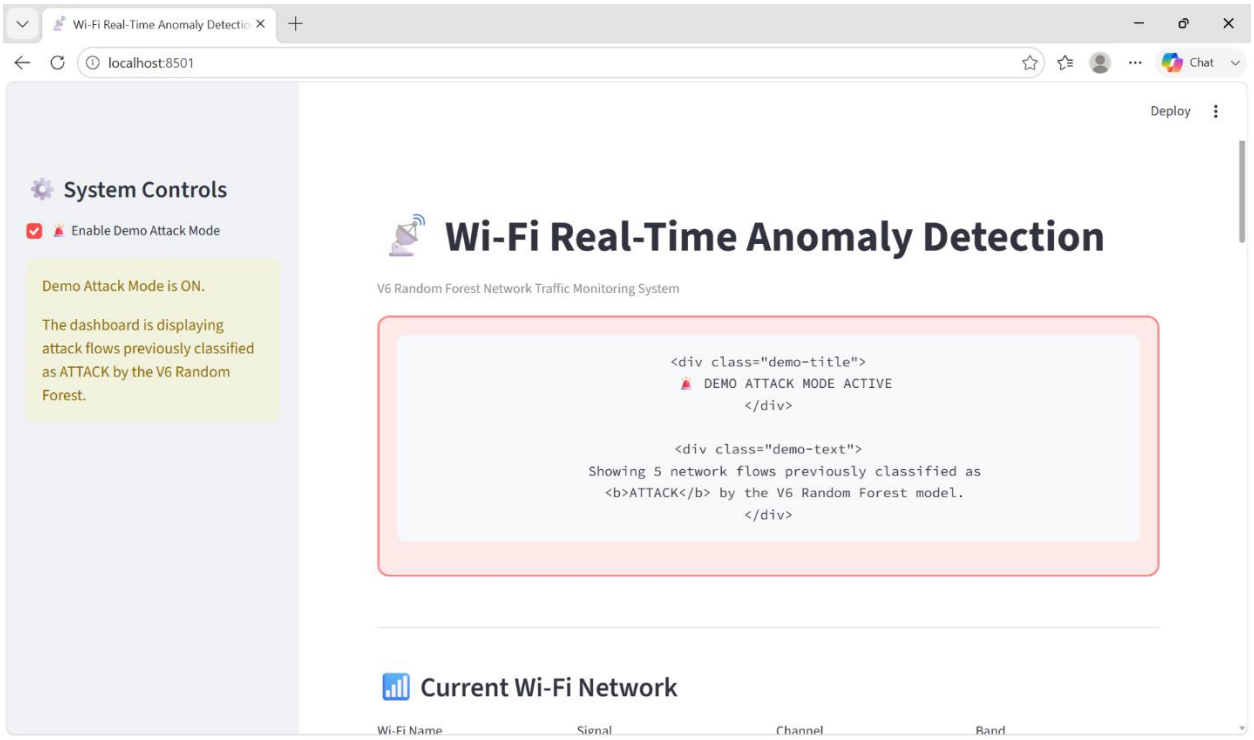
8.1.5 Final Result



8.1.6 Result When Detection Stopped



8.1.7 Demo Attack Mode



8.2 Sample Output Summary

The developed Real-Time Wi-Fi Network Traffic Anomaly Detection System using the V6 Random Forest model produces output through the Streamlit-based dashboard. The system continuously captures Wi-Fi network packets, processes them into network flows, extracts 68 traffic features, applies the V6 preprocessing and scaling steps, and classifies each flow as BENIGN or ATTACK. The prediction result and confidence value are stored in the detection result file and displayed on the dashboard.

The dashboard provides a real-time summary of the monitored network, including the connected Wi-Fi information, total number of detected flows, benign traffic, detected attacks, and anomaly rate. It also presents traffic sources based on source IP, destination IP, destination port, and protocol. The detailed traffic section allows the user to examine individual flow records along with their prediction results and confidence values.

CONCLUSION

The Real-Time Wi-Fi Network Traffic Anomaly Detection System using the V6 Random Forest model was successfully designed and implemented to monitor Wi-Fi network traffic and identify potentially anomalous activities in real time. The system integrates live packet capture, flow processing, feature extraction, data preprocessing, machine learning classification, result storage, and dashboard visualization into a unified monitoring framework. The implemented system captures network packets from the Wi-Fi interface and organizes them into network flows. From these flows, 68 traffic features are extracted and processed using the V6 preprocessing and scaling approach. The processed features are then provided to the V6 Random Forest model, which classifies network flows as BENIGN or ATTACK and provides a prediction confidence value. The detection results are stored for further analysis and are presented through the Streamlit-based dashboard. The dashboard provides a clear view of live network activity through traffic statistics, traffic sources, detailed flow information, and detected anomalies.

Future Enhancements:

1. **Device Identification:** Extend the system to identify connected devices and monitor their traffic patterns, where network access and permissions allow.
2. **Improved Detection Accuracy:** Train the Isolation Forest model using larger, real-world WiFi traffic datasets to reduce false positives and improve anomaly detection.
3. **Advanced Machine Learning Algorithms:** Compare Isolation Forest with algorithms such as Local Outlier Factor (LOF) and One-Class SVM to improve detection performance.
4. **Cloud Integration:** Store traffic logs and anomaly reports in a secure cloud database to support remote monitoring and long-term analysis.
5. **Adaptive Learning:** Update the detection model periodically to adapt to changing network conditions and newly observed traffic patterns.

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CYBER SECURITY, MACHINE LEARNING ALGORITHMS

Real-Time Wi-Fi Network Traffic Anomaly Detection Using a Supervised Random Forest Model

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Abstract

The Wi-Fi Real-Time Anomaly Detection System is a machine-learning-based network security solution designed to monitor network traffic and identify abnormal or potentially malicious activities in real time. The system captures live network packets using Npcap and processes them using Scapy to construct network flows. From each flow, 68 network-flow features are extracted, representing characteristics such as packet counts, packet lengths, flow duration, inter-arrival times, protocol flags, and transmission rates. The extracted features are preprocessed using the same scaling and transformation pipeline applied during model development and are then provided to a Random Forest classifier for traffic classification. The system is developed using the CIC-IDS2017 dataset, which contains both benign and attack traffic. The training and evaluation data represent several attack categories, including PortScan, DDoS, DoS, FTP-Patator, SSH-Patator, Bot activity, Heartbleed, and web attacks. The V6 model classifies observed traffic into BENIGN or ATTACK categories. For real-time monitoring, the prediction results are continuously stored and displayed through a Streamlit dashboard, which provides Wi-Fi information, traffic statistics, source-wise summaries, live network flows, prediction confidence, and anomaly alerts. The proposed system can be applied in companies, colleges, and other organizational networks to continuously monitor traffic and provide an early indication of suspicious network behaviour. The system also supports controlled testing using labelled attack flows to demonstrate its anomaly-detection capability.

Keywords: Wi-Fi security; anomaly detection; Random Forest; network traffic monitoring

1 Introduction

The implementation integrates real-time packet capture, flow construction, 68-feature extraction, preprocessing, Random Forest classification, and Streamlit visualization for continuous Wi-Fi anomaly detection. Our earlier study investigated Wi-Fi traffic anomaly detection using the Isolation Forest approach [1]. Building upon that work, the present study focuses on the implementation and evaluation of the developed network traffic anomaly detection pipeline. The present implementation incorporates packet capture using Npcap and Scapy, flow construction, a fixed 68-feature representation, stored preprocessing, supervised classification, real-time logging, and a Streamlit-based monitoring interface. Isolation Forest is attractive for a lightweight prototype because it is an unsupervised method based on recursive random partitioning. The original method showed that anomalous observations tend to be isolated with shorter path lengths [5].

2 Related work

2.1 Wireless conditions and the base paper

The principal base paper, published in IEEE Access in 2025, investigates dynamic spectrum access in dense Wi-Fi environments using deep reinforcement learning. It models multiple access points and heterogeneous traffic in an NS-3 environment and uses environmental observations such as interference, traffic load, delay, and signal-related information for dynamic channel decisions [2].

2.2 Network anomaly detection

Tao et al. proposed a parallel network-traffic anomaly detection method based on Isolation Forest and Spark for large-scale traffic processing [3].

2.3 Adaptive wireless IDS

IDS-Twin moves wireless intrusion detection toward streaming and self-adaptive anomaly modelling using a digital-twin-based replay environment and contrastive learning [4].

3 Implementation

3.1 Packet capture and traffic collection

The Phase-2 system was implemented in a Windows environment for real-time observation of Wi-Fi network traffic. Npcap was used to capture packets from the selected wireless interface, while Scapy was used for packet processing. The captured traffic was continuously monitored and converted into flow-level information for subsequent analysis.

3.2 Flow construction

Captured packets were grouped into network flows instead of being analysed individually. Each flow records communication characteristics such as source and destination information, ports, transport protocol, duration, packet count, and related traffic statistics. This flow-based representation enables temporal and directional properties of network communication to be considered during detection.

3.3 Feature extraction and preprocessing

Each constructed flow was converted into a fixed **68-feature representation**. The features include packet and byte counts, packet-length statistics, inter-arrival-time characteristics, transport flags, directional rates, subflow statistics, and active/idle measurements. The extracted feature vector was passed through the stored preprocessing pipeline before classification.

3.4 Machine learning classification

The preprocessed flow features were provided to the trained Random Forest classifier, which generated a binary BENIGN or ATTACK decision along with its confidence. The binary classification provides a direct indication of whether the observed flow resembles the traffic patterns learned during model training.

4 Results and performance evaluation

4.1 Experimental setup

The developed Phase-2 system was evaluated using a supervised machine-learning approach for network-flow classification. Unlike an unsupervised anomaly scoring approach, the V6 model was trained to distinguish normal network behaviour from traffic patterns associated with malicious activity. The final V6 feature set consists of 68 numerical network-flow attributes. These attributes describe multiple characteristics of a

communication flow, including packet counts, packet lengths, flow duration, inter-arrival times, transmission rates, TCP flags, header information, window sizes, and active/idle behaviour. Before classification, the selected features are subjected to the same preprocessing pipeline used during model development. Logarithmic transformation is applied to appropriate traffic-rate features, followed by feature scaling. The resulting feature vector is then supplied to the trained Random Forest classifier.

For independent evaluation, a separate Tuesday traffic dataset containing 445,909 records was used. The dataset consisted of 432,074 BENIGN flows, 7,938 FTP-Patator flows, and 5,897 SSH-Patator flows. No feature mismatch occurred during testing, as all 68 features expected by the V6 model were available in the independent dataset.

4.2 Classification performance

The proposed Random Forest classifier achieved the performance shown in Table 1 on the independent Tuesday test data.

Table 1. Performance of the proposed Random Forest classifier on the independent Tuesday test data.

Metric	Proposed system performance
Accuracy	99.99%
Precision	99.65%
Recall	100.00%
F1-Score	99.82%

The high recall indicates that the evaluated attack flows were successfully identified by the classifier. The precision value also indicates that only a small number of normal flows were incorrectly classified as malicious. The overall accuracy was approximately 99.99%, demonstrating strong classification performance on the independent evaluation set.

4.3 Confusion matrix analysis

Table 2 gives the binary confusion matrix obtained in the Tuesday evaluation.

Table 2. Binary confusion matrix for the Tuesday evaluation.

Actual / Predicted	BENIGN	ATTACK
BENIGN	432,025	49
ATTACK	0	13,835

The classifier correctly identified 432,025 benign flows and 13,835 attack flows. Only 49 benign flows were incorrectly classified as attacks, while no attack flows in this test set were classified as benign.

4.4 Attack-wise detection performance

The independent test also enabled evaluation of the two attack classes present in the Tuesday test set (Table 3).

Table 3. Attack-wise detection performance on the Tuesday test set.

Attack type	Total samples	Detected	Detection Rate
FTP-Patator	7,938	7,938	100%
SSH-Patator	5,897	5,897	100%

Both FTP-Patator and SSH-Patator traffic were successfully detected in the independent evaluation. The result demonstrates that the learned network-flow patterns were effective for the attack categories represented in this test set.

4.5 Dashboard output

A real-time monitoring dashboard was developed using Streamlit to display the active Wi-Fi network information, including SSID, signal strength, channel, frequency band, and transmission rates, together with live network traffic statistics and anomaly detection results.

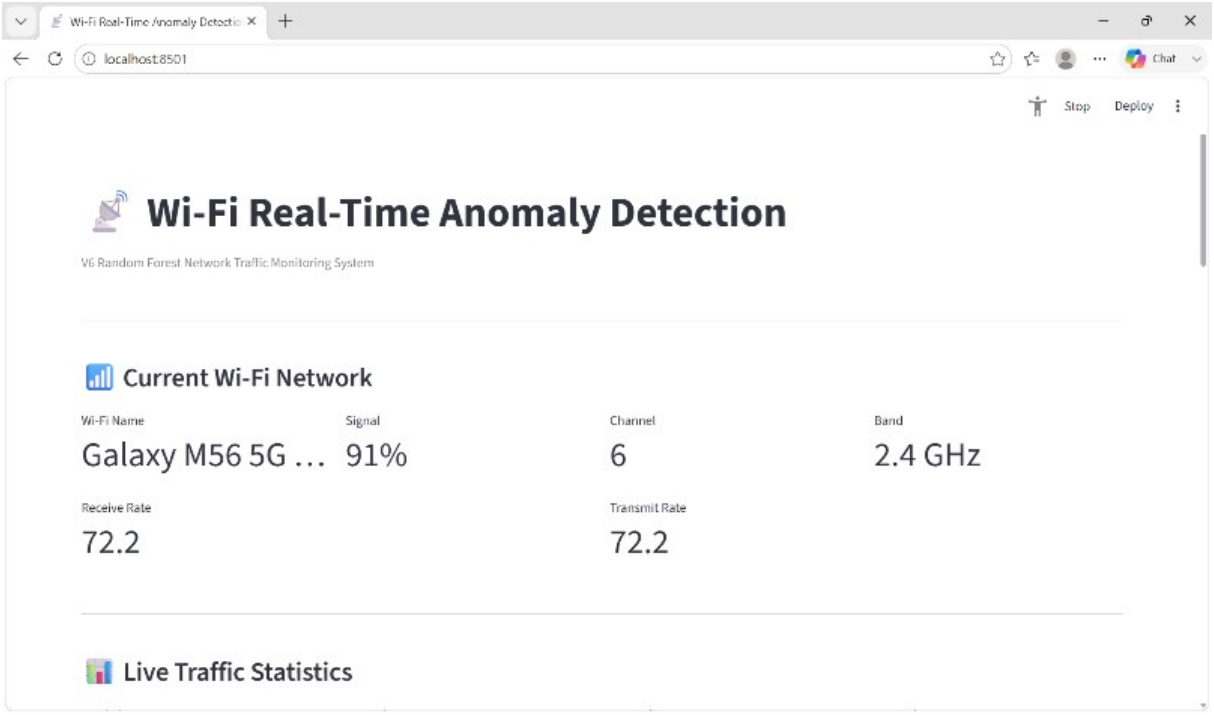


Figure 1. Real-time Wi-Fi network monitoring and anomaly detection dashboard.

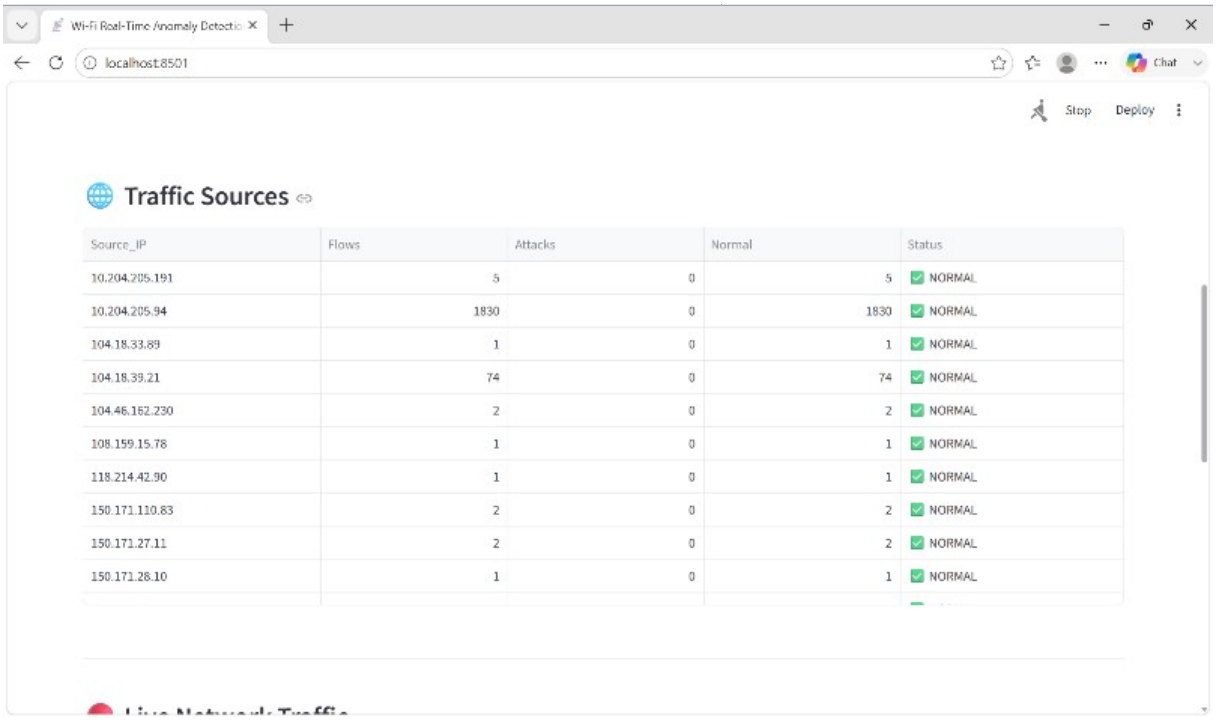


Figure 2. IP addresses communicating with the monitored laptop.

Table 4. Fields of the source-wise summary of IP addresses communicating with the monitored laptop and their observed flows.

Field	Meaning
Source_IP	IP address involved in the observed traffic
Flows	Number of network flows associated with that IP
Attacks	Number of those flows classified as attacks
Normal	Number classified as normal
Status	Overall status based on the detection results

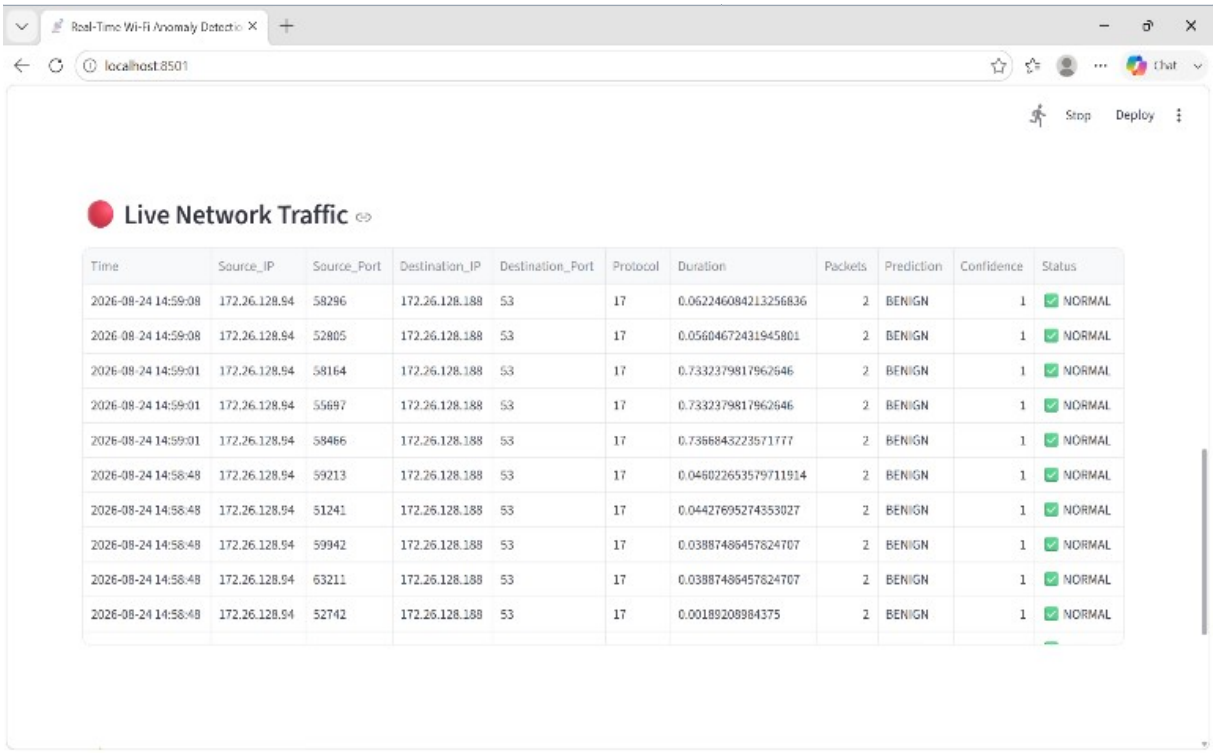


Figure 3. Live network traffic section of the dashboard.

Table 5. Columns of the live network traffic view, showing what the back end detects in real time.

Column	Meaning
Time	When the network flow was observed
Source_IP	Device/source generating the traffic
Source_Port	Port used by the source
Destination_IP	Remote/local destination
Destination_Port	Destination service port
Protocol	17 = UDP, 6 = TCP
Duration	Duration of that flow in seconds
Packets	Number of packets in the flow
Prediction	Our model's classification
Confidence	Model confidence
Status	Human-readable result: NORMAL or ATTACK

4.6 Flow features

The main flow features are computed as follows:

$$\text{Flow Duration} = t_n - t_1 \tag{1}$$

$$\text{Total Fwd Packets} = N_f \tag{2}$$

$$\text{Total Bwd Packets} = N_b \tag{3}$$

where N_f and N_b are the numbers of packets travelling in the forward and backward directions. The total forward and backward bytes are

$$\text{Total Length of Fwd Packets} = \sum_{i=1}^{N_f} L_i \tag{4}$$

$$\text{Total Length of Bwd Packets} = \sum_{j=1}^{N_b} L_j \tag{5}$$

where L_i (L_j) is the length of the ith forward (jth backward) packet, and t_1 and t_n are the times of the first and last packets of the flow.

5 Conclusion

The proposed system provides an effective approach for real-time network anomaly detection using 68 flow-level features and a Random Forest classifier. The system successfully combines live traffic monitoring with machine-learning-based classification and a user-friendly dashboard. The results demonstrate its potential for practical network-security monitoring. Future work can focus on detecting additional attack types and improving real-time deployment across larger networks.

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